



安徽理工大学

ANHUI UNIVERSITY OF SCIENCE & TECHNOLOGY

计算机科学与工程学院
本科毕业设计（论文）附件材料

基于图像增强的恶劣天气车辆感知系统设计

中文题目：	计与实现		
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2026 年 5 月

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文献翻译

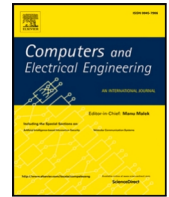
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2026 年 5 月



A novel approach to low-light image and video enhancement using adaptive dual super-resolution generative adversarial networks and top-hat filtering

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ARTICLE INFO

Keywords:

Adaptive fusion
Generative adversarial network
Image and video enhancement
Mobile devices
Super-resolution
Top-hat transform

ABSTRACT

Image and video enhancement under low-light conditions is challenging, as the task involves more than just brightness adjustment. Without addressing issues such as artifacts, distortions, and noise in dark regions, brightness improvement alone can worsen the quality. This paper presents a novel approach to low-light image and video enhancement based on the adaptive fusion of Dual Super-Resolution Generative Adversarial Network (DSRGAN) models, followed by Top-Hat Gradient-Domain Filtering (THGDF). A soft thresholding mechanism is used to integrate the Memory Residual Super-Resolution Generative Adversarial Network (MRSRGAN) and the Weighted Perception Super-Resolution Generative Adversarial Network (WPSRGAN). MRSRGAN enhances fine details, improving the objective performance of the image, while WPSRGAN improves overall details, enhancing the subjective performance. Top-hat gradient-domain filtering is then applied to remove artifacts, distortions, and noise in both images and videos, resulting in outstanding perception scores. The proposed approach is validated using the quality assessment metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measurement (SSIM), and Information Fidelity Criterion (IFC). Extensive experiments conducted on publicly available source codes and databases demonstrate that the proposed method is more effective than the existing state-of-the-art techniques.

1. Introduction

Capturing clear and detailed images or videos in low-light conditions is challenging. Due to inadequate lighting, noise levels increase, degrading the overall image quality. Additionally, long exposure times used to gather more light can introduce motion blur, affecting the sharpness of the captured image. The reduced dynamic range also results in overexposed highlights and underexposed shadows. These issues have been effectively addressed by numerous researchers using deep learning models [1–3], video stabilization techniques [4], video restoration methods [5], and more. Although, these approaches yield improved results, video enhancement under extreme low-light conditions with artifacts, distortions and noise remains a challenge.

Existing research has introduced promising model-based schemes [6] and data-driven methods [1] to enhance images and videos. Conventional mathematical approaches can improve resolution, contrast, brightness, and color under moderate lighting conditions. However, enhancing the quality of an image and video under low-light conditions, especially in the presence of noise, remains

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<https://doi.org/10.1016/j.compeleceng.2024.110052>

Received 31 July 2024; Received in revised form 15 December 2024; Accepted 29 December 2024

Available online 13 January 2025

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an area for improvement. Image and video enhancement, particularly reconstruction under noisy conditions, continues to be a challenge. These enhancement techniques improve the viewing experience both visually and informatively through operations such as temporal filtering, super-resolution, color correction, contrast adjustment, deblurring, and noise reduction. Image and video enhancement techniques are widely used across various scientific and commercial applications. Low-light enhancement has a wide range of applications, including video surveillance [7], autonomous driving [8], medical imaging [9], and computational photography [10].

Smartphone users face significant challenges in improving low-light images and videos due to limited hardware capabilities, processing power, and high user expectations. Limited aperture sizes on mobile devices result in insufficient light capture, making low-light enhancement even more difficult. Real-time processing requirements, along with the limited computing power of smartphones, necessitate efficient algorithms that balance performance with speed. Memory constraints also limit the complexity and size of models that can be employed for these enhancements. Hence, algorithm optimization is crucial to meet user expectations. The need for seamless image and video enhancement under low-light conditions is especially immediate requirement for the smartphone users. Balancing technical challenges with the demand for high-quality performance is critical for improving mobile phone stored images and videos in low-light conditions.

The primary goal of low-light image enhancement (LLIE) is to improve the interpretability and perception of images captured in low-light conditions. In recent years, deep learning-based solutions incorporating various learning strategies, network architectures, loss functions, and training data have gained significant attention in this domain. Advances in deep learning and computer vision have led to the continuous development of sophisticated video enhancement approaches. It is likely that these methods will soon be implemented via user-friendly web interfaces and unified online platforms, which will include various popular LLIE techniques. Research in this area is increasingly important as efforts continue to address the challenges of low-light enhancement for smartphone applications.

Conventional low-light image and video enhancement methods include histogram equalization (HE) and retinex-based models. Among these, retinex-based methods have gained the most attention in recent research. Many researchers have contributed to developing retinex-based deep learning models [11,12]. Retinex-based approaches typically decompose low-light images into reflection and illumination components using priors and regularization techniques, with the enhanced result being the estimated reflection component. However, retinex-based methods have limitations. In complex lighting conditions or extremely low-light situations, color balance and detail enhancement are often compromised. Additionally, these methods can introduce artifacts or unrealistic enhancements that need to be addressed in future LLIE developments to create more robust solutions.

One of the most popular techniques for restoring images under low-light conditions is single image super-resolution (SISR) [13]. Super-resolution (SR) algorithms reconstruct HR images from corresponding low-resolution (LR) images. Deep learning algorithms typically perform faster and yield better results than traditional SR algorithms, as demonstrated by various performance metrics such as PSNR, SSIM, and IFC. Deep learning-based SR algorithms excel at visual enhancement. According to the literature, SR algorithms can be divided into two categories. In the first category, Convolutional Neural Networks (CNNs) are used as optimization functions with traditional L1 or L2 loss functions in pixel space. In the second category, Generative Adversarial Networks (GANs) are employed, introducing a novel approach to image reconstruction.

Lintao et al. [14] introduced a novel framework for low-light image enhancement that synthesizes normal and low-light image pairs to enhance training data. By employing an illumination degradation model and a self-attention-based enhancement model, the approach improves robustness and performance across various tasks, outperforming existing methods. It effectively augments data, demonstrates superior performance, and improves robustness in complex scenes. However, it lacks discussion on computational complexity and the results are not satisfactory with generalization in highly varied environments. Kumar et al. [15] presents the Morphological Transformation-based Locality-Constrained Representation (MTLcR) model for low-light robust face SR. This approach detects low-light effects in input images using domain-specific dictionary images, applying similar effects to LR dictionary images to minimize reconstruction errors. The incorporation of morphological transformations into the objective function enhances reconstruction weights, resulting in improved performance. Simulation results on publicly available datasets and real-world images demonstrate the model's superior reconstruction capability under low-light conditions compared to state-of-the-art methods. However, the paper lacks with a detailed analysis of computational complexity and scalability in diverse scenarios.

1.1. Contributions

This paper presents DSRGAN, a novel SR framework for low light image and video enhancement. The main contributions of the proposed method work as mentioned below:

a. **Objective** The primary objective of this paper is to address the challenges of enhancing image and video quality under low-light conditions. The authors recognize that merely adjusting brightness is insufficient; the enhancement process must also tackle artifacts, distortions, and noise prevalent in dark regions. This work proposes a novel method that utilizes advanced GANs and Top-Hat Gradient-Domain Filtering techniques to improve the perceptual and objective quality of images and videos captured in low-light environments.

b. **Data Collection and Preprocessing** In this work, the sources of images and videos are collected from five publicly available benchmarks for image SR: Sun-Hays, Urban100, Set5, Set14, and a mobile dataset. The Sun-Hays, Urban100, Set5, and Set14 datasets primarily consist of natural images, while the mobile dataset contains images captured from various device models, including the iPhone 3GS, BlackBerry Passport, Sony Xperia Z, and Canon 70D DSLR, all of which are freely available

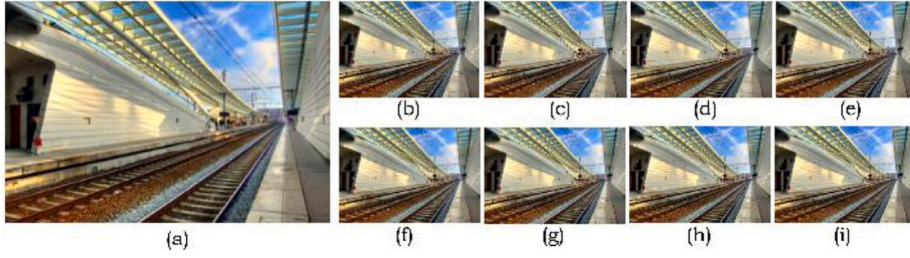


Fig. 1. An illustration of railway station image reconstructed using existing methods and its comparison with the proposed method. (a) Low Resolution Input Image (PSNR:22.87, SSIM:0.6, IFC:1.8) (b) Ground Truth Image (c) BGAN [16] (23.47, 0.67, 1.89), (d) RGSR [17] (23.45, 0.65, 1.76), (e) BDRVSR [18] (23.51, 0.61, 1.81), (f) RESRGAN [19] (23.56, 0.64, 1.79), (g) DBVSR [20] (23.66, 0.68, 1.82), (h) BlindSR [21] (23.39, 0.60, 1.85), and (i) Proposed Method (23.7, 0.69, 1.98).

for public download. Preprocessing steps include normalization of pixel values, data augmentation through techniques like rotation and cropping, and the generation of LR versions of HR images, ensuring the input data is suitable for effective training and evaluation of the proposed methodologies.

- c. **Methodology** The methodology consists of an adaptive fusion approach combining DSRGAN with THGDF. The DSRGAN models are designed to enhance fine and overall details in the images. A soft thresholding mechanism is employed to integrate the outputs of the MRSRGAN, which focuses on detail enhancement, and the WPSRGAN, which aims to improve overall visual quality. The THGDF is then applied to address and eliminate any residual artifacts, distortions, and noise, ensuring a visually appealing output.
- d. **Feature Learning** The feature learning of the model includes specific settings to effectively train and evaluate models. The training dataset comprised Flickr2K, with 2,650 HR images, while evaluation datasets included Urban100, Sun-Hays, Set5, and Set14, providing over 100 images for assessment. The LR images are generated using bicubic downsampling, with a LR patch size of 24×24 pixels and an HR patch size of 96×96 pixels. Data augmentation techniques, including random horizontal and vertical flipping, enhanced model robustness, with a batch size set to 16. For the MRSRGAN, the generator architecture featured 7 residual blocks with a filter size of 3×3 pixels. The learning rate was initialized at 1×10^{-4} using the Adam optimizer, trained over 500 epochs. In the WPSRGAN, the generator had 16 residual blocks and utilized a PReLU activation function.
- e. **Evaluation Metrics** The performance of the proposed method is assessed using standard quality assessment metrics, including PSNR, SSIM, and IFC. These metrics provide quantitative measures of image quality and enable comparison with existing methods. The paper presents detailed evaluations demonstrating how the proposed technique achieves higher scores in these metrics, further validating its effectiveness in low-light image and video enhancement.
- f. **Results** Numerous experiments were conducted to evaluate the performance of the proposed approach against existing state-of-the-art techniques. The benchmark methods used are BGAN [16], RGSR [17], BDRVSR [18], RESRGAN [19], DBVSR [20], and BlindSR [21]. The experimental results presented demonstrate that the proposed method is more effective compared to other state-of-the-art methods.

Fig. 1(a) shows a low-light image generated from the ground truth or high-resolution (HR) image. Fig. 1(b) shows the HR image. Figs. 1 (c) through (h) depict results from other SR methods such as BGAN [16], RGSR [17], BDRVSR [18], RESRGAN [19], DBVSR [20], and BlindSR [21]. The result of the proposed method is shown in Fig. 1(i). As it is evident from the PSNR, SSIM, and IFC metric comparisons, the proposed method reconstructs images more effectively than other existing techniques.

The rest of the paper is organized as follows: Section 2 reviews the existing work in four stages, outlining the contributions made by other researchers. Section 3 details the proposed approach. Section 4 presents the experimental results of the proposed method on standard datasets. The results demonstrate that the proposed method produces more visually appealing, vivid, and brilliant images and videos compared to other competitive methods. Although the results presented focus on images, the proposed method has also been tested on standard video datasets. Finally, Section 5 summarizes the key contributions of this work.

2. Background and related works

Several SISR algorithms based on deep learning have been proposed by various researchers, showing significant performance improvements [22,23]. This paper focuses on five key aspects in the literature review: SR using Convolutional Neural Networks (SRCNN) [24], Very Deep SR (VDSR) [25], Enhanced Deep SR (EDSR) [26], SR using GANs (SRGAN) [27], and image quality evaluation.

2.1. Super-resolution convolutional neural network (SRCNN)

SRCNN is a standard in image SR that demonstrates how deep learning can be applied to enhance image quality. Dong et al. [24] introduced SRCNN, which is designed to directly learn the complex mapping between LR and HR images. Feature extraction is

accomplished through three convolutional layers. Unlike other artificial intelligence (AI) techniques, SRCNN operates on image patches, allowing it to focus on local attributes within the data. As a supervised learning algorithm, SRCNN optimizes its parameters using paired LR and HR images to minimize the difference between predicted and ground truth images. SRCNN offers several compelling advantages compared to other existing methods. Firstly, its architecture is designed for simplicity while achieving superior accuracy compared to state-of-the-art example-based methods. Secondly, it employs a moderate number of filters and layers, enabling SRCNN to operate efficiently in practical online applications, even on a CPU. SRCNN outperforms a series of example-based methods due to its fully feed-forward nature, eliminating the need to solve optimization problems during runtime. Thirdly, experimental work of other investigation shows that enhancing the network's restoration quality is feasible with larger datasets and/or a larger model. In contrast, larger datasets or models can pose challenges for existing example-based methods.

The initial step in SRCNN involves up-scaling a single LR image to the desired size using bi-cubic interpolation. This serves as the pre-processing step. The resulting interpolated image is denoted as Y . The objective of SRCNN is to recover from Y an image $F(Y)$ that closely resembles the ground truth HR image. For clarity, Y is referred to as an LR image, even though its size matches that of the HR image X . The model proposed by Lu et al. [28], based on shuffle learning, is designed for deployment on mobile devices with limited computing power. Through extensive experiments on three widely used datasets, Lu's model not only achieves high PSNR and SSIM but is also easier to implement compared to other image super-resolution (SR) models due to its simplicity.

Many applications require HR images because they include complementary information. However, improving pixel density through sensor and visual technologies is limited and costly. Elgohary et al. [29] used image processing methods to generate HR images from their LR counterparts to overcome this challenge. Video Super-Resolution (VSR) aims to recover complex details and mitigate sensor effects. The analysis utilizes wavelet analysis and CNNs to combine multi-frame SR with wavelet analysis, resulting in a multiscale representation of video frames obtained through wavelet decomposition. Consequently, several CNNs are trained independently to approximate wavelet multiscale characteristics. Inference on these trained CNNs is conducted through regression of wavelet multiscale representations from LR frames. Additionally, wavelet reconstruction is performed, resulting in reconstructed HR frames. The research comprises the use of various camera zoom lenses to capture the LR frames and create HR frames. The authors conducted several experiments demonstrating the effectiveness of their approach in restoring video frames.

The review of existing work shows that CNNs have demonstrated exceptional performance in SR techniques. However, these models may not be feasible for low-cost devices. The Multiscale Feature Selection Network (MFSN), introduced by Minghon et al. [30], addresses this challenge. The Multiscale Feature Selection Block (MFSB), at the core of MFSN, is developed to extract rich multiscale features from coarse-to-fine receptive fields. In each section of MFSB, excluding the last, a wide-activated residual unit is employed to enhance representation capability. By adaptively varying the receptive field sizes of two adjacent divisions within MFSB, the Scale Selection Module (SSM) efficiently merges features from these branches. Furthermore, a Comprehensive Channel Attention Mechanism (CCAM) is seamlessly integrated into SSM, dynamically adapting selection weights based on local and global inter-channel dependencies. The results presented by the authors show that MFSN compares favorably with other lightweight methods. SR is inherently a multi-solution process due to its one-to-many mapping from LR to HR space. However, existing SR networks treat the SR of different sizes for the same image as independent tasks, making it challenging to leverage feature learning experiences and requiring extensive computational time to recover HR images. These issues also exist in current arbitrary-scale SR methods.

Wenyuan et al. [31] introduced a novel self-generating (SG) multiscale learning method based on SG mechanisms to achieve efficient and effective HR image recovery. The SG-SR method uses feature learning outcomes from SR networks to generate upscale filters as an alternative to traditional upscale modules. As each scale factor is adjusted, the SG upscale module provides weights to filter the LR tensor, resulting in HR images corresponding to the filtered tensors. Based on numerous experiments and comparisons of benchmark datasets, Wenyuan's method outperforms state-of-the-art methods. Additionally, the SG upscale module significantly reduces computational costs in association with other upscale modules while enhancing the effectiveness of existing SR networks.

By integrating explicit feature learning into CNNs, Qizhou et al. [32] enhance the representation capabilities of CNNs through Trainable Feature Matching (TFM). The TFM features can be explicitly learned from training images through feature matching by incorporating trainable feature sets. Qizhou's Trainable Feature Matching Attention Network (TFMAN) improves SR performance by incorporating non-local and channel attention. Due to computational demands, the authors introduce a streamlined variant called Same-size-divided Region-level Non-Local (SRNL), which performs non-local computations in parallel on uniformly divided blocks from the input feature map. The effectiveness of SRNL and TFM is validated through ablation studies, with comprehensive experimental results from benchmark datasets showing that TFMAN consistently outperforms other methods.

The review of existing methods shows that SRCNNs are prevalent in image SR. CNN structures can be adaptively modified, and advanced self-attention mechanisms can be incorporated to enhance SR performance. A common characteristic among these improvements is that CNNs implicitly learn feature processing modes resembling convolution and non-local operations instead of explicitly focusing on high-frequency details. Conversely, a dictionary-based approach explicitly learns feature decompositions to match and reconstruct LR features (see Table 1).

2.2. Very deep super-resolution (VDSR)

Through deep CNNs, Very Deep Super-Resolution (VDSR) advances the boundaries of resolution enhancement in image SR. VDSR, initially developed by Kim et al. [25] in 2016, which incorporates a deeper network design to capture more complex and subtle features inherent in HR images. VDSR can learn complex mappings between LR and HR image spaces by leveraging significantly

Table 1
Summary of SRCNN Methods.

Method	Key Features	Strengths	Limitations
SRCNN [24]	Uses deep learning to directly map LR to HR images through 3 convolutional layers. Performs image patch extraction.	Simple architecture with high accuracy. Efficient for online applications, even on CPU.	Requires paired LR-HR images. Performance can improve with larger datasets and models, which may not always be feasible.
SISR [28]	Model designed for mobile devices with limited computing power. Simple architecture with high PSNR and SSIM.	Easy implementation with high performance on standard datasets.	May not scale well to larger or more complex datasets.
WCNN VSR [29]	Combines wavelet analysis with CNNs for multi-frame SR. Performs wavelet decomposition and reconstruction for HR video frames.	Effective for video SR and mitigating sensor effects.	Computationally intensive for video data, and requires specialized camera equipment for capturing LR frames.
MFSN [30]	Features a Multiscale Feature Selection Block (MFSB) and a Comprehensive Channel Attention Mechanism (CCAM) for extracting rich features.	Lightweight and effective for low-cost devices. Dynamically adjusts receptive field sizes and merges features efficiently.	May not be as effective for very high-resolution images or large-scale datasets.
SG-SR [31]	Uses self-generating multiscale learning to create upscale filters, reducing computational costs while enhancing SR network performance.	Significantly reduces computational cost while maintaining SR effectiveness. Outperforms existing methods in terms of efficiency and performance.	May be less flexible in handling highly varied image types or real-time applications.
TfMAN & SRNL [32]	Incorporates Trainable Feature Matching (TFM) and non-local channel attention to improve SR performance. Introduces a streamlined version (SRNL) for parallel computation.	Enhances feature matching with non-local attention, improving SR performance. Streamlined for computational efficiency.	Computationally demanding for larger datasets. The effectiveness of SRNL is still reliant on the quality of the feature extraction process.

increased convolutional layers. With residual learning, VDSR streamlines the optimization process. VDSR enhances performance by learning residuals, the difference between LR inputs and their corresponding HR targets.

VDSR achieves notable improvements in SR tasks, enabling the reconstruction of fine details and textures in HR images due to its deep architecture and residual learning principles. In addition to learning complex feature representations through multiple layers of abstraction, VDSR effectively captures complex image characteristics. VDSR can understand complex feature representations through various layers of abstraction. Further, VDSR's usage and popularity highlights the importance of architectural innovations in deep learning. VDSR demonstrates how deeper networks can unlock new performance levels for various computer vision tasks. Through deep learning methodologies, VDSR represents a fundamental step forward in pursuing super-resolution images. It was observed in Fig. 2 that in all cases the network trained with depth-wise fine-tuning converges more quickly, which support our hypothesis. Notably, both the speed of convergence and the restoration performance have been substantially improved. This improvement is especially evident with deeper neural networks, indicating the potential for depth-wise fine-tuning to enhance training stability.

Diffusion CNNs (DCNNs) are widely used to extract features and reconstruct details for SISR. However, many existing DCNN methods cannot fully leverage the complementary strengths of feature maps, channels, and pixels. This limits their ability to represent abundant features. Therefore, the specified problem is efficiently addressed using the Weidong et al. [33] Cascaded Visual Attention Network (CVANet) method. CVANet focuses on reconstructing details and is inspired by the visual attention mechanism of the human eyes. CVANet comprises a trainable Feature Attention Module (FAM) for learning feature-level attention, a channel attention module for improving feature maps by learning channel-level attention, and a pixel attention module for generating HR images. This is accomplished by adaptively selecting representative features from preceding layers. CVANet consistently outperforms state-of-the-art methods regarding subjective visual perception, PSNR, and SSIM on four benchmarks. The authors have provided CVANet's code using the following URL: <https://github.com/WilyZhao8/CVANet>.

Medical imaging, especially fluorescence and ultrasound captures, requires advanced solutions to track molecular dynamics accurately in real-time in cellular environments. In this context, deep learning-based image denoising offers significant improvements over conventional methods. The Multiscale Self-Attention Network (MSAN) is a new architecture proposed by Kyungsu et al. [34] for optimal denoising of fluorescence and ultrasound images. An MSAN consists of a feature extraction layer to identify high and

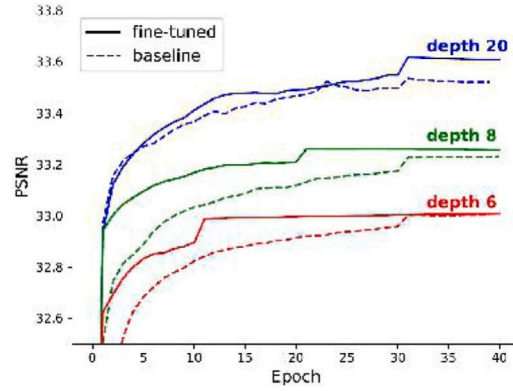


Fig. 2. VDSR results at various depths. Dotted lines represent the baseline models, while solid lines indicates the depth-wise fine-tuned networks.

low-frequency attributes, a multiscale self-attention mechanism for predicting residuals from original and downsampled feature maps, and a residual image decoder. A denoised image is obtained by offsetting the residual output. Compared to state-of-the-art models such as RIDNet [35] and DnCNNp [36], MSAN achieves notable PSNR improvements for the Fluorescence Microscopy Denoising (FMD) [37], W2S [38], and ultrasound datasets. It is clear from the specified method that MSAN effectively improves denoising capabilities for fluorescence and ultrasound images in real-time, enabling the decoding of complex biological processes.

Yeh et al. [39] aim to ease temporal flickering when upscaling videos by Fine-Grained VSR (FGVSR). However, several existing methods need help to utilize spatial-temporal information effectively from LR videos, resulting in unwanted artifacts or inaccurate reconstructions of the image's details. To overcome these issues, the author proposed a deep learning framework for FGVSR, which takes consecutive LR video frames as input and generates SR frames corresponding to those frames. The framework emphasizes the alignment and refinement of multiple frames to reconstruct missing information from LR sources. An alignment module is introduced to align numerous frames at the feature level, preventing flickering in the video output. A feature fusion module is used once the aligned features have been fused and refined. On top of that, a refinement module is presented to reconstruct missing information based on the fused features. An image enhancement module is embedded in the skip connection for further refinement. The authors showed that deep FGVSR outperforms existing deep learning-based VSR methods. Detailed experiments are performed on REDS, Vid4, and Vimeo90k benchmarks, achieving state-of-the-art performance. In comparing FGVSR and state-of-the-art VSR methods, FGVSR shows that PSNR has substantially improved.

With the rise of deformable alignment-based VSR methods, VSR has gained significant traction in creating HR videos from LR counterparts. SR quality can be enhanced by aligning features extracted from videos to correct motion errors. Existing SR methods often suffer from challenges such as misalignment and insufficient temporal information, resulting in artifacts. Zhu et al. [40] efficiently addressed the mentioned challenges based on the Deep VSR Network (DVSRNet). This approach integrates the Progressive Deformable Alignment (PDA) module and the Temporal-Sparse Enhancement (TSE) module. By propagating bidirectional information, the PDA module precisely aligns features and eliminates artifacts; furthermore, the TSE module enhances HR frames.

Chen et al. [41] proposed a lightweight deep optical flow network to obtain bidirectional optical flows for the PDA module. A new loss function is also designed to generate HR frames with sharp, precise details. The results presented based on the demonstration shows that the method is superior over the state-of-the-art approaches. This approach emphasizes its ability to overcome misalignment issues and improve the overall quality of VSR. Existing approaches based on deep learning methods commonly use LR optical flow for VSR, in which HR frames are generated from LR frames. However, LR optical flow may have difficulty recovering fine details. This work combines an optical flow SR and an optical flow enhancement algorithm to enhance temporal dependency accurately. A supplementary information component is extracted from adjacent LR frames for precise feature extraction. An optical flow enhancement algorithm is used to infer HR optical flow from LR. Motion compensation is then applied to the HR optical flows. As a second benefit, extracting the detail component mitigates the errors introduced by motion compensation based on optical flow. An SR network is then used to generate SR results. A comprehensive set of experiments on two datasets, Vid4 and DAVIS, demonstrate that the VSR method outperforms other state-of-the-art techniques to achieve high-quality SR (see Table 2).

2.3. Enhanced deep super-resolution (EDSR)

Enhanced Deep Super-Resolution (EDSR) is a cutting-edge advancement in image SR. The EDSR offers an improved way to reconstruct HR details from LR inputs. Lim et al. [26] proposed EDSR architecture in 2017 as shown in Fig. 3 and introduced key improvements to achieve optimal performance based on VDSR. By embracing a profound architecture, the EDSR can learn increasingly complex representations of image features, surpassing its predecessors in the number of layers. As a result of the incorporation of residual blocks, the optimization process is further simplified. The network can now learn the residuals between LR and HR images, resulting in highly detailed and realistic HR outputs from the model.

Table 2
Summary of VDSR Methods.

Method	Key Features	Strengths	Limitations
VDSR [25]	Uses a deep CNN with residual learning to improve SR performance by learning residuals between LR and HR images.	Achieves fine detail reconstruction through deep networks. Faster convergence with depth-wise fine-tuning.	Requires large training datasets and substantial computational resources.
CVANet [33]	Utilizes cascaded visual attention mechanisms (feature, channel, and pixel attention modules) for image reconstruction.	Outperforms state-of-the-art methods on PSNR, SSIM, and subjective perception.	Computationally intensive, requiring high memory and processing power.
MSAN [34]	Combines multiscale self-attention and residual learning to enhance denoising performance for fluorescence and ultrasound images.	Achieves high PSNR for medical imaging denoising, particularly in real-time.	May not be generalizable to non-medical image denoising tasks.
FGVSR [39]	Uses temporal refinement and alignment techniques for improving VSR, preventing flickering and enhancing detail.	Outperforms existing VSR methods with significant PSNR improvement.	Computationally demanding, especially for large video datasets.
DVSRNet [40]	Integrates Progressive Deformable Alignment and Temporal-Sparse Enhancement modules to improve video SR.	Effective in overcoming misalignment issues and improving VSR quality.	Requires sophisticated optical flow algorithms and can be computationally expensive.
Optical Flow VSR [41]	Combines optical flow enhancement with deep SR networks to generate high-quality HR video frames.	Superior to conventional methods in restoring fine details and handling temporal dependency.	Dependent on high-quality optical flow estimates; may struggle with complex motion patterns.

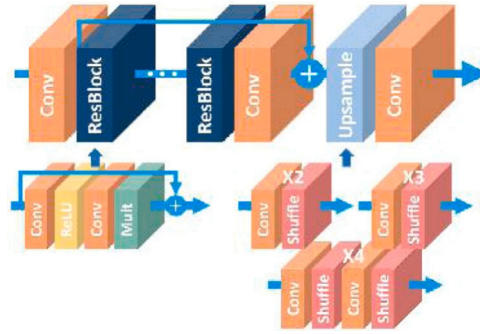


Fig. 3. EDSR Architecture.

With its streamlined and simplified architecture, EDSR distinguishes itself not only by its depth but also by its simplicity. The EDSR has consistently achieved top-tier results in super-resolution challenges through extensive experiments and rigorous assessments. With its ability to capture fine details and produce visually appealing results, EDSR has become a benchmark in the field. The EDSR inspires researchers to explore the potential of increasingly sophisticated deep-learning architectures. EDSR influences subsequent developments in image super-resolution. By leveraging deep neural networks, EDSR demonstrates the solution to complex mappings that primarily improve images.

2.4. Super-resolution generative adversarial network (SRGAN)

A breakthrough in image super-resolution can be attributed to the Super-Resolution Generative Adversarial Network (SRGAN). The SRGAN incorporates the power of generative adversarial networks (GANs). Ledig et al. [27] introduced SRGAN in 2017 as a new paradigm incorporating adversarial training. SRGAN consists of a generator network that generates SR and a discriminator network that distinguishes between HR images. Using this adversarial training framework, the Generator produces HR outputs aligned with

ground truth images and perceptual quality. A more comprehensive evaluation of SR performance is possible with SRGAN. The SRGAN emphasizes the subjective visual realism of the generated images. It transcends traditional metrics such as PSNR and SSIM by emphasizing the subjective visual realism of the generated images.

In the Internet of Things (IoT) context, intelligent cameras are increasingly being used to perform quality inspection tasks. However, low-light conditions may pose operational challenges for these smart-cameras. This situation severely degrades the acquired images, resulting in a decline in the performance of the subsequent detection algorithms. Jing et al. [42] proposed a Low-light image Enhancement GAN (LEGAN) for industrial internet of smart-cameras model for the industrial internet of intelligent cameras, focusing on non-stationary noise compression and detail recovery. The input is decomposed into two branches, of which the first step uses Harr-wavelets. These branches are encoded independently using compact residual blocks, facilitating effective noise suppression. The second step incorporates a feature selection module to extract correlations between image foregrounds and backgrounds.

Further, the reconstruction is accomplished based on comprehensive feature map correlations. In the training phase, an optimal balance between noise suppression and detail recovery is ensured by an adversarial loss comprising MSE loss, VGG loss, and discriminating loss. It is evident from the results presented that the LEGAN performs better than other noise compression and detail recovery schemes. In industrial practice, a high detection accuracy was achieved in low-light environments, specifically for yarn evenness inspection.

In SISR, an HR image is reconstructed from a LR image. This ill-posed problem arises because one LR image maps to multiple HR possibilities. It has been demonstrated that generative models-based SISR methods outperform traditional approaches in terms of performance. However, GANs, Variational Auto-Encoders (VAEs), and Flow face challenges such as unstable training, low sample quality, and high computational costs. As a result, these models need help to achieve the trifecta of diversity, high quality, and fast sampling. Despite their impressive sample variety and quality, denoising diffusion probabilistic models do not apply to real-world situations due to their prohibitive sampling costs. To overcome this limitation, Xiao et al. [43] proposed SISR with Denoising Diffusion GANs (SRDDGAN), which aims for large step denoising, diverse samples, and stable training results. SRDDGAN utilizes multimodal conditional GANs for each denoising step, combining denoising diffusion models with GANs. The results demonstrate that the SRDDGAN outperforms existing diffusion model-based methods regarding PSNR and perceptual quality metrics. The latent variable Z solution significantly improves HR spatial domain diversity. In addition, SRDDGAN infers nearly 11 times faster than diffusion-based SR3, making it more practical for real-world applications.

A retinal fundus image is an essential diagnostic tool in ophthalmology that facilitates the early detection and monitoring of various ocular diseases. It is challenging to capture HR fundus images due to factors such as defocusing, diffraction, limited shutter speed, sensor unit density, and random noise. Super-resolution techniques can overcome this limitation, which enhances visual details in retinal fundus images. The traditional SR methods tend to introduce artifacts into texture details and compromise delicate retinal vessel structures. Liquan et al. [44] proposed GANs that consist of a generator and a discriminator to improve the perceptual quality of retinal fundus images. A generator comprises 23 multiscale feature extraction blocks, an image segmentation network, and 23 residual-in-residual dense blocks. These components are strategically used to extract retinal vessel grayscale images and vascular features. It consists of two branches, one for extracting global features and another for extracting vascular features.

Further, the extracted features are fused to restore the SR image as best as possible. This approach can restore retinal vessel images in more detail and accurately. Additionally, attention modules are added to the Discriminator to improve the quality of SR images. An artifact loss function is added to the GAN to enhance the ability to measure the disparity between the HR and restored images. Compared with state-of-the-art methods of image super-resolution, the images generated by this method exhibit better perceptual quality.

Guo et al. [45] proposed a crucial aspect of maintaining high-quality spatial resolution in remote sensing images by preserving image texture during super-resolution reconstruction. The deep learning super-resolution methods based on GANs successfully recover image texture. This study aims to improve the SR reconstruction of medium and LR remote sensing images using an enhanced GAN. Enhancements to the residual block and refinement of the model loss function are introduced in the network, which builds upon the SRGAN. The model loss function seeks to minimize the pixel distance between the reconstructed remote sensing and the original image. In addition, the network restores texture details based on semantic categories when perceptual loss is targeted. This method outperforms SRGAN over other networks in both subjective and objective assessments. Based on the results presented, this approach generates more realistic and reliable textures, improving the objective quantitative evaluation of the reconstructed images. Kabiraj et al. [46] examine LR images of number plates where poor boundary contrast and accuracy make it challenging to detect numbers. To improve detection and recognition accuracy, low-quality number plate images must be digitized correctly, and image resolution needs to be enhanced. This approach is based on enhanced SRGAN, which modifies native dense blocks with a residual dense block model. An activation layer based on Rectified Linear Units (ReLU) complements CNNs for threshold. Enhanced plate images are segmented using an optical character recognition (OCR) model to detect and recognize characters on the number plates. The OCR model achieves an average accuracy of 84% for HR images, while for LR images, it varies from 4% to 7%. As the resolution of the plate images increases, the model accuracy increases, demonstrating its effectiveness. According to the OCR model, Enhanced Super-Resolution with Generative Adversarial Network (ESRGAN) provides superior LR image enhancement compared to SRGAN and Pro-SRGAN methods. Converting LR images to HR ones improves digit or alphabet detection accuracy on number plates. This approach improves number plate recognition reliability in LR situations.

Yu et al. [47] address the challenges of obtaining clear images of particleboard surfaces. This essential forest product saves forestry resources and promotes low-carbon development by reusing wood processing residues. The large size of particle boards poses difficulties in capturing detailed image features, resulting in blurred defect outlines. To enhance the quality of particleboard

Table 3
Summary of SRGAN Methods.

Method	Key Features	Strengths	Limitations
SRGAN [27]	GAN-based architecture for high-quality image super-resolution. Uses adversarial training with generator and discriminator.	Realistic high-resolution images, improved perceptual quality.	Training instability, computationally intensive.
LEGAN [42]	Focuses on low-light image enhancement for smart cameras. Incorporates noise suppression and detail recovery.	Effective for industrial inspection, improves detection in low-light.	May not generalize to other use cases.
SRDDGAN [43]	Combines denoising diffusion models with GANs for stable and fast training.	Faster training, higher perceptual quality compared to diffusion models.	Computationally complex, struggles with extreme noise.
GAN for Retinal Fundus Images [44]	Uses generator-discriminator approach for improving perceptual quality, with multiscale extraction blocks and attention modules.	High-quality SR for medical images, especially vascular features.	Requires large datasets, may not generalize to other medical areas.
Enhanced SRGAN for Remote Sensing [45]	Enhances SRGAN with residual blocks and refined loss function for better texture preservation.	Superior texture restoration, higher evaluation scores.	May not perform well in dynamic environments.
ESRGAN for Number Plate Recognition [46]	Enhanced SRGAN with residual dense blocks for improving LR number plate images, integrates OCR for detection.	Increases accuracy of number plate recognition.	Limited to number plate recognition applications.
SRDAGAN for Particleboard Surface Images [47]	Optimized SRGAN using dense skip connections and CBAM for improving particleboard images.	Outperforms traditional methods in texture clarity.	Computationally demanding, not suited for low-resource environments.
Dual-Path GAN for Satellite Images [48]	Dual-path architecture combining feature mapping attention with deep convolution for satellite SR.	Enhanced clarity, feature extraction for satellite images.	Complex architecture, high computational requirements.

surface images, the study employs SR reconstruction technology, focusing on improving the SRGAN performance. The presented enhancement method is named SR dense attention GAN (SRDAGAN). This method refines the SRGAN model by removing the batch normalization (BN) layer.

Further, the Convolutional Block Attention Module (CBAM) is optimized for constructing dense blocks. In addition, dense skip connections are incorporated into the dense block construction. It is clear from the results presented that the training, verification, and testing conducted on 52,400 image blocks with HR and LR in a 3:1:1 ratio produces better results. Comprehensive evaluation metrics, such as PSNR, SSIM, and Learned Perceptual Image Patch Similarity (LPIPS), demonstrate the superiority of SRDAGAN over traditional methods such as BICUBIC, SRGAN, and SWINIR. The reconstructed images have more evident textures and provide a realistic representation of limitless features, indicating a significant enhancement in the model performance. The study discovers the reconstruction effect on images with defects, emphasizing that SRDAGAN successfully achieves high-quality SR reconstruction of particleboard images.

Ren et al. [48] address the hardware limitations, complex electromagnetic noise, and harsh natural environment challenges associated with the quality of satellite remote-sensing images. To enhance the clarity of these images, the study proposes a dual-path adversarial generation network algorithm specifically designed for satellite remote sensing image SR. The generator structure incorporates a dual-path convolution operation, combining a feature mapping attention mechanism with an enhanced deep convolutional network. The design uses LR images to extract both important features and deep features. A reconstruction layer optimizes the SR process by fusing the extracted data. Improvements are made to the algorithm's loss function and Discriminator to better balance the output image and the Discriminator. The fine-tuning aims to bring the SR image closer to human perception. As evident from the results presented compared to other methods, this algorithm performs significantly better on the UCAS-AOD datasets <https://hyper.ai/datasets/5419> (see Table 3).

2.5. Image quality evaluation

It has become increasingly important to evaluate enhanced image quality as image super-resolution (SR) techniques continue to advance. To assess the performance of SR algorithms effectively, robust metrics must consider both objective and subjective aspects of image quality. Objective evaluation methods, such as PSNR, SSIM, and IFC [49] are widely used;

a. Peak Signal to Noise Ratio

The PSNR is a widely used metric for evaluating image quality, particularly in the context of lossy image compression and reconstruction techniques such as super-resolution. The PSNR is expressed in decibels (dB) and is calculated as Eq. (1):

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\text{MSE}} \right) \quad (1)$$

where L is the maximum possible pixel value of the image, and MSE is the Mean Squared Error between the original image and the reconstructed image, defined as Eq. (2):

$$\text{MSE} = \frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2 \quad (2)$$

In this formula $I(i, j)$ represents the pixel value at position (i, j) in the original image. $K(i, j)$ represents the pixel value at position (i, j) in the reconstructed image, m and n are the dimensions of the images. Combining these, the PSNR formula can be written as shown in Eq. (3):

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{L^2}{\frac{1}{mn} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2} \right) \quad (3)$$

- b. **Structural Similarity Index Measurement (SSIM)** The next metric used to assess the SR methods is the SSIM. The SSIM is a method for measuring the similarity between two images. It considers changes in structural information, luminance, and contrast to assess image quality. SSIM is widely used because it provides a more comprehensive evaluation of image quality compared to metrics such as MSE and PSNR. The SSIM index between two images x and y is defined as Eq. (4):

$$\text{SSIM}(x, y) = \frac{(2 \mu_x \mu_y + C_1)(2 \sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \quad (4)$$

where μ_x is the average of image x , μ_y is the average of image y , σ_x^2 represent the variance of image x , σ_y^2 is the variance of

image y , σ_{xy} is the covariance of images x and y , C_1 is given by $(K_1 L)^2$ and C_2 is given by $(K_2 L)^2$

In the above equations L is the dynamic range of the pixel values (e.g., 255 for 8-bit grayscale images). K_1 and K_2 are small constants with the typical values, $K_1 = 0.01$ and $K_2 = 0.03$.

- c. **Information Fidelity Criterion (IFC)** The IFC metric is used to assess the quality of an image by quantifying the amount of information shared between a reference (original) image and a distorted (reconstructed) image. It is based on the concept of natural scene statistics and the notion that the human visual system is highly adapted to the statistical properties of natural scenes. The IFC is derived from the mutual information between the reference image and the distorted image. It can be expressed as shown in Eq. (5):

$$\text{IFC}(I, K) = \sum_{j=1}^N \log_2 \left(1 + \frac{2 x_j y_j}{\frac{2}{x_j} \frac{2}{y_j} - \frac{2}{x_j y_j}} \right) \quad (5)$$

where I is the reference image, K is the distorted image, x_j and y_j are the coefficients of the reference and distorted images, respectively, in a multi-scale, multi-orientation image decomposition (e.g., wavelet transform), $\frac{2}{x_j}$ is the variance of the reference image coefficients, $\frac{2}{y_j}$ is the variance of the distorted image coefficients, $\frac{2}{x_j y_j}$ is the covariance of the reference and distorted image coefficients, and N is the number of coefficients.

Despite of objective evaluation metrics presented, there remains a gap in objective evaluation criteria that align closely with human subjective perception. In the following section, the Perceptual Index (PI) and Root Mean Square Error (RMSE) metrics that are based on human perception of image quality are presented.

- d. **Perceptual Index (PI)** The PI is an image quality metric that combines the Ma score and the Naturalness Image Quality Evaluator (NIQE) to approximate human perception of image quality. This approach is especially valuable in evaluating images that have been enhanced or generated, as it accounts for both aesthetic appeal and naturalness. The Perceptual Index is calculated using Eq. (6)

$$\text{Perceptual Index, PI} = \frac{1}{2} (10 - \text{Ma} + \text{NIQE}) \quad (6)$$

where the Ma score measures the aesthetic quality of the image, with higher scores indicating greater aesthetic appeal, and NIQE evaluates the naturalness of the image, where lower NIQE scores suggest a more natural appearance. Together, these scores provide a balanced assessment of how visually pleasing an image may be to a human observer.

A lower Perceptual Index score suggests a higher perceived image quality, reflecting both a high Ma score (indicating visually pleasing or attractive to human observers) and a low NIQE score (indicates naturalness). Conversely, a higher PI score typically indicates a less visually satisfying image. This metric is particularly useful in image enhancement tasks such as low-light image enhancement, super-resolution, and de-noising, where human perception of quality is essential. In applications like GANs for image synthesis, the Perceptual Index can help assess the realism of generated images. Moreover, it is helpful for comparing different image enhancement algorithms, providing a perceptually relevant comparison that aligns with human satisfaction.

- e. **Root Mean Square Error (RMSE)** RMSE is a commonly used metric to evaluate the performance of regression models in machine learning. It quantifies the average magnitude of errors between predicted values and actual (observed) values. RMSE is calculated by first finding the difference between each predicted value and its corresponding actual value, then squaring these differences to remove negative signs and give more weight to larger errors. These squared errors are averaged, and then the square root of this average is taken, yielding the RMSE value. Mathematically, RMSE is given by Eq. (7):

$$\text{Root Mean Square Error, RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (7)$$

where y_i represents the actual values, $\hat{y}_i$ represents the predicted values, and n is the total number of observations.

RMSE is particularly useful in regression since it is sensitive to larger errors and provides a metric in the same units as the target variable, making it easy to interpret. In machine learning, RMSE is commonly used for model evaluation: a lower RMSE indicates a model that better fits the data. RMSE also aids in model comparison, where the model with the lowest RMSE is typically considered the best. Furthermore, in hyperparameter tuning, RMSE often serves as the metric to minimize when optimizing model parameters, such as through grid or random search, to achieve better accuracy.

The evaluation schemes offer a comprehensive understanding of the effectiveness of image SR techniques in enhancing visual appeal and perceptual accuracy.

3. Proposed method

It is clear from the introduction section that low-light image and video enhancement present complex challenges beyond simple brightness adjustment. In low-light environments, issues such as artifacts, color distortion and noise are common. Techniques are needed to ensure accurate color representation, remove artifacts, and to apply appropriate filters for noise reduction. Increasing brightness in such settings is also challenging due to noise amplification. Therefore, methods that reduce noise while preserving visual clarity are crucial. Additionally, fine details can be lost due to independent image processing, making it necessary to balance global enhancements for improved visibility with the preservation of local details. Achieving this balance requires addressing both subjective and objective performance metrics, considering diverse scenes, and adapting to real-time processing constraints. The integration of enhancement techniques, user preferences, and metrics based validation further complicates the development of effective low-light image and video enhancement methods.

The proposed method incorporates an adaptive strategy, advanced algorithms, and comprehensive assessment metrics to address the identified challenges. The framework, Dual SRGAN, adaptively fuses the MRSRGAN and WPSRGAN networks. Fig. 4 presents the flow diagram of the proposed architecture. The perception scores [50] and RMSE are used to achieve an optimal balance between global enhancement and local detail preservation. The MRSRGAN improves objective performance by enhancing fine details, while subjective performance is improved by the WPSRGAN. A thresholding mechanism is used to adaptively fuse the MRSRGAN and WPSRGAN networks.

Additionally, a top-hat gradient-domain filter is employed to remove noise and artifacts resulting from SR. PSNR, SSIM, and IFC metrics are used to validate the results of the proposed method. The experimental results presented demonstrate the effectiveness of the proposed approach. As shown in the results, the proposed method quantitatively outperforms current state-of-the-art techniques. Under low-light conditions, it can produce vivid, clear, and brilliant images and videos that closely resemble human visual system perception.

The Sections 3.1, 3.2, and 3.3 introduces the architectures of the MRSRGAN and WPSRGAN networks, followed by their adaptive fusion process. Further, the overall architecture of the Dual SRGAN framework is presented.

3.1. Architecture of MRSRGAN

MRSRGAN enhances fine details in images and videos by utilizing Memory Residual (MR) blocks. The memory residual blocks store and leverage information from previous layers to emphasize and refine the complex features. Through training, the network improves its ability to capture and highlight delicate structures under low-light conditions by learning and retaining critical details.

Fig. 5 presents the complex architecture of MRSRGAN, which is explicitly designed for SR tasks. A single convolutional layer extracts features from LR images, serving as input for both the Generator and Discriminator of the MRSRGAN. Image quality is enhanced through the integration of seven MR blocks to the architecture of MRSRGAN. The integrated MR blocks improve the model's ability to retain important features by creating persistent memory. By incorporating residual connections, the model effectively captures long-range dependencies, emphasizing the differences between LR and HR features. The Generator refines these features iteratively using residual blocks, followed by up-sampling blocks. These up-sampling blocks increases the spatial resolution of images and videos. Finally, the HR images are generated through the last convolutional layer, demonstrating the network's capability.

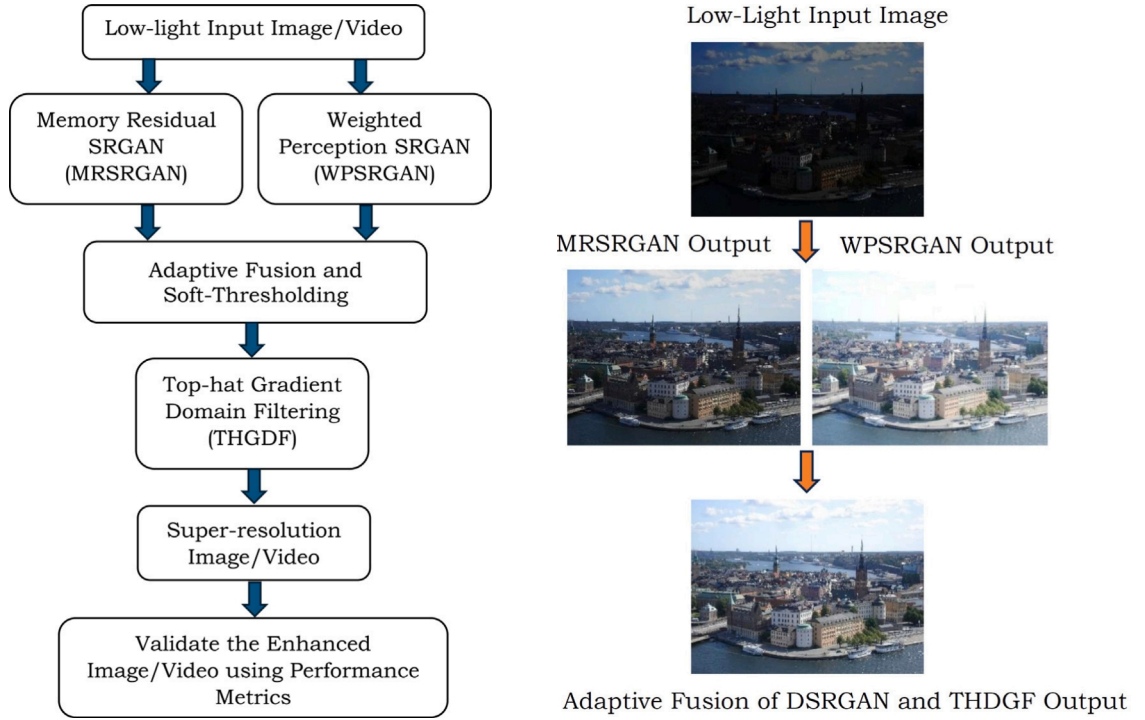


Fig. 4. (a) Flow Diagram of the Proposed Architecture (b) Image Transformation of Adaptive Dual SRGAN.

A sophisticated mechanism unfolds within each MR block. Each MR block consists of four Residual Blocks (ResBlocks) and a gate unit. After generating these groups, the gate unit extracts a specified number of features. Combining the input features with the extracted features yields the final output of the MR block. The ResBlocks are transformed using Parametric Rectified Linear Unit (PReLU) functions, which replace the activation function layers. To simplify the overall architecture, the BN layers are excluded from the computation. Two up-sampling operations are necessary to restore the original image size. Each convolutional layer is assigned a stride, with parameters represented as n indicating the number of corresponding feature maps. This configuration optimizes the generator network's computational efficiency while ensuring that the original image's essential structural and functional aspects are preserved during SR processing. The discriminator network employs the same settings as the generator network.

3.1.1. Loss function

The total generator loss function, as defined by Eq. (8), can be decomposed into pixel-wise, adversarial, and perceptual loss components.

$$\text{Total Loss, } L_{\text{total}} = L_{\text{pixel}} + L_{\text{adv}} + 2L_{\text{vgg}} \quad (8)$$

In the defined loss formulation, three distinct components contribute to the overall objective function for the MRSRGAN: pixel-wise Mean Squared Error (MSE) loss (L_{pixel}), perceptual loss based on the VGG19 network features (L_{vgg}), and the adversarial loss for the Generator (L_{adv}). Each term serves a specific purpose in guiding the training of the generator network to produce high-quality SR images.

- a. **Pixel-wise MSE Loss (L_{pixel}):** The L_{pixel} represented by Eq. (9) quantifies the pixel-wise mean squared error between the generated images and the ground truth. Here, $G(x_l)$ denotes the Generator that produces the SR images, while x_l and x_t represents the LR input images and their corresponding ground truth, respectively. This loss term focuses on minimizing the discrepancy at the pixel level, aiding the Generator in producing images that closely resemble the ground truth in terms of raw pixel values.

$$\text{Pixel-wise MSE Loss, } L_{\text{pixel}} = \frac{1}{N} \sum_{i=1}^N |x_l^i - G(x_l^i)|^2 \quad (9)$$

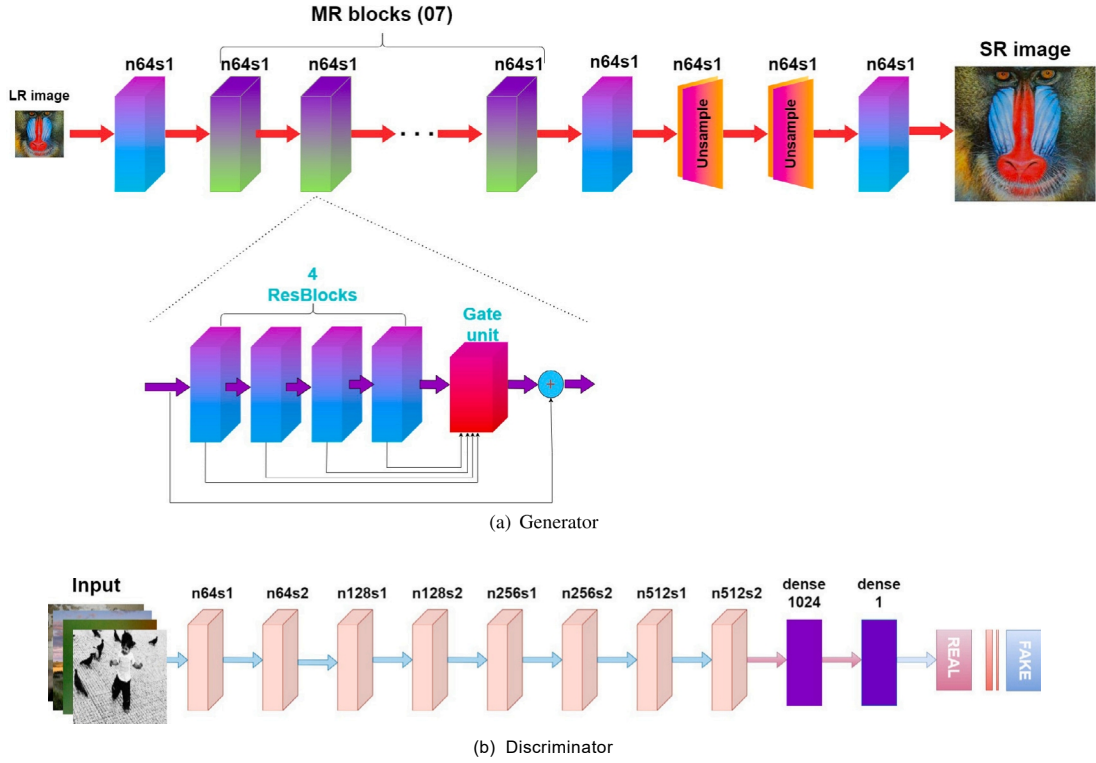


Fig. 5. The Memory Residual SRGAN Architecture (a) Generator, (b) Discriminator.

- b. **Perceptual Loss** (L_{vgg}): The L_{vgg} loss defined by Eq. (10) introduces perceptual loss, calculated as the MSE loss between features extracted from the pre-trained VGG19 network. By utilizing features from a deep CNN like VGG19, the Generator is driven to produce images that not only match the ground truth at the pixel level but also capture higher-level semantic information. This enhances the perceptual quality of the generated images. The number of training samples is indicated as N . The term x_t^i represents the features extracted from the pre-trained VGG19 network.

$$\text{Perceptual Loss, } L_{vgg} = \frac{1}{N} \sum_{i=1}^N |x_t^i - G(x_t^i)|^2 \quad (10)$$

- c. **Adversarial Loss** (L_{adv}): The L_{adv} loss, defined by Eq. (11) represents the adversarial loss for the Generator, a fundamental component of the GAN framework. In this context, the logarithm is omitted from the adversarial loss calculation. Adversarial loss helps the Generator to produce images that appear homogeneous when evaluated by the Discriminator. The weights associated with adversarial and perceptual losses are denoted by λ_1 and λ_2 , respectively. The combined loss function integrates pixel-wise accuracy, perceptual quality, and adversarial training to guide the MRSRGAN in generating high-fidelity super-resolved images. The weights λ_1 and λ_2 allow for fine-tuning the contributions of adversarial and perceptual losses during the training process.

$$\text{Adversarial Loss, } L_{adv} = -D(G(x_t)) \quad (11)$$

3.2. WPSRGAN architecture

WPSRGAN improves overall image enhancement by incorporating a weighted perceptual loss. This approach enhances subjective performance by prioritizing overall visual quality and global features. The network assigns different levels of importance to various perceptual features, ensuring a balanced and enhanced global appearance.

The WPSRGAN architecture, comprising 16 ResBlocks within the generator network, is shown in Fig. 6. In the Generator, each ResBlock consists of a convolutional layer, a PReLU activation layer, and a second convolutional layer. WPSRGAN deliberately omits BN layers from both the generator and discriminator components. The detailed structure of each ResBlock in the Generator is outlined as follows:

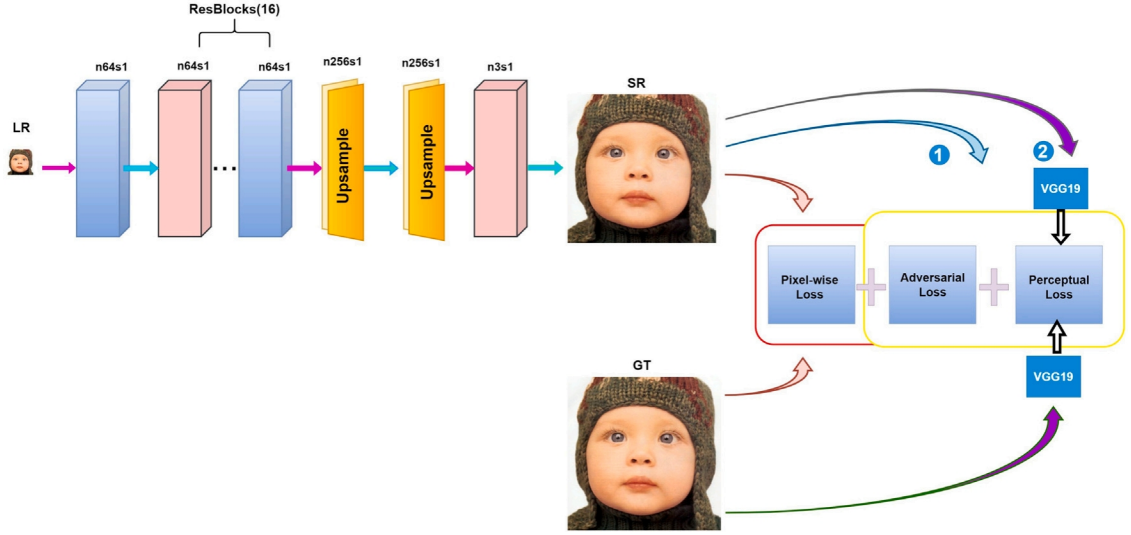


Fig. 6. The Generator Architecture of WPSRGAN.

Convolutional Layer (First): The ResBlock begins with a convolutional layer that extracts features from the input data. This layer performs spatial convolution to capture essential information relevant to the super-resolution task.

PReLU Activation Layer: The convolutional layer is followed by a PReLU activation layer. PReLU provides a non-linear activation function, enabling the network to model complex relationships and capture complex details in the data.

Convolutional Layer (Second): A second convolutional layer refines the features extracted by the preceding layers. This dual-convolutional structure within each ResBlock allows for more effective capture and representation of hierarchical features.

The decision to remove BN layers from both the Generator and Discriminator of WPSRGAN was made to simplify the architecture and reduce computational complexity. BN layers can sometimes introduce unwanted artifacts or hinder training stability, which led to their exclusion. The Discriminator in WPSRGAN is similar to that of MRSRGAN, with the key distinction being the absence of BN layers. The Discriminator distinguishes between authentic and generated images, and this architectural adjustment maintains consistency between the two models while emphasizing the removal of BN layers.

The WPSRGAN network architecture consists of 16 ResBlocks in the Generator, each comprising a convolutional layer, a PReLU activation layer, and another convolutional layer. The absence of BN layers in both the Generator and Discriminator aligns with the weighted perceptual loss approach and underscores the emphasis on simplicity in design.

Loss Function: As depicted in Fig. 6, WPSRGAN employs a two-stage bias adversarial training mechanism utilizing distinct generator losses. During the initial stage, highlighted, the generator loss is optimized, comprising pixel-wise loss and adversarial loss, to enhance the overall objective performance (specifically, to minimize the RMSE value). In the subsequent stage, the network parameters from the first stage is treated as a pre-trained model. Subsequently, generator loss is replaced with perceptual and adversarial loss to refine and optimize subjective visual effects (e.g., minimizing the perceptual index). The two-stage losses are defined by Eqs. (12) and (13).

$$L_1 = L_{\text{pixel}} + \lambda_1 L_{\text{adv}} \quad (12)$$

$$L_2 = \lambda_1 L_{\text{adv}} + \lambda_2 L_{\text{vgg}} \quad (13)$$

Here, the pixel-wise loss is defined as Eq. (9), the perceptual loss utilizes MSE loss calculated based on features extracted from a pre-trained VGG19 network, and the adversarial loss is represented as Eq. (14),

$$L_{\text{adv}} = -\log(D(G(x))) \quad (14)$$

Implementing a two-stage adversarial training mechanism facilitates aligning the generated SR image with the corresponding ground truth in the high-level features space.

3.3. Adaptive fusion

The adaptive fusion approach employed in the proposed framework maximizes the strength of both networks by combining MRSRGAN and WPSRGAN outputs. To align the perceptual score with RMSE in SR results, a soft thresholding method is applied.

The network outputs are then merged using a dynamic mechanism, where the threshold is adjusted based on specific characteristics of the image or video inputs.

A distinct SR outcomes are derived using the two previously mentioned GANs. One is MRSRGAN, prioritizing enhanced objective performance. The other is WPSRGAN, which achieves results with superior subjective perception. To reunite the perceptual score, the soft thresholding method proposed by Deng et al. [13] is adopted. In order to merge the two SR images (i.e., MRSRGAN, WPSRGAN), a technique similar to a form of pixel interpolation is adopted. Eq. (15) presents the soft thresholding method adopted in the proposed work:

$$I_e = I_G + \text{soft}(\cdot; \cdot) \quad (15)$$

where I_e is the fused image and $\cdot = (I_G - I_{\mathcal{G}})$, I_G is the generated image by WPSRGAN whose perceptual score is lower, $I_{\mathcal{G}}$ is the generated image by MRSRGAN whose RMSE value is lower. $\cdot$ is the adjusted threshold. The $\text{soft}(\cdot; \cdot)$ is defined in Eq. (16)

$$\text{soft}(\cdot; \cdot) = \text{sign}(\cdot) \cdot \max(|\cdot| - \cdot, 0) \quad (16)$$

3.4. Dual SRGAN (DSRGAN) framework

The adaptive fusion approach presented in the earlier section integrates MRSRGAN and WPSRGAN networks to arrive at the Dual SRGAN (DSRGAN) network. This approach leverages the specific strengths of each network to enhance low-light images and videos comprehensively. A DSRGAN is based on the fusion of MRSRGAN and WPSRGAN results. DSRGAN represents a unified network that integrates the enhanced fine details of MRSRGAN and the globally improved perceptual quality of WPSRGAN. Integrating local and global aspects of low-light content is crucial.

3.5. Top-hat gradient domain filtering (THGDF)

Dual SRGAN outputs are subjected to top-hat gradient-domain filtering to remove image artifacts, noise and balance global enhancement and local detail preservation. The filtering process refines the enhanced results to effectively address low-light conditions and other challenges while maintaining high visual fidelity.

3.5.1. Top-hat transformation

The top-hat transformation [51] is the most widely used operation in mathematical morphology. It is used for image enhancement which involves adding bright areas while subtracting dark areas from the original image. Enhancing the performance of the top-hat transform is often employed within a multiscale framework. This approach allows the multiscale top-hat transform to extract various beneficial features from the image, subsequently utilized for infrared image enhancement. Widely adopted across diverse grayscale image types, the multiscale top-hat transform scheme has demonstrated efficacy in enhancing retinal, ultrasound, and infrared images. Furthermore, it finds application in tasks such as visible and infrared image fusion, image segmentation, and small object detection.

The top-hat transform constitutes a composite operation within mathematical morphology derived from other fundamental morphological operations, including erosion, dilation, opening, and closing. The morphological operations of dilation and erosion applied to $I(u,v)$ for $B(s,t)$, denoted as $I \boxplus B$ and $I \ominus B$ are defined as shown in Eqs. (17) and (18):

$$(I \boxplus B)(u; v) = \max_{(s,t) \in I} \{I(u+s; v+t) + B(s; t)\} \quad (17)$$

$$(I \ominus B)(u; v) = \min_{(s,t) \in B} \{I(u+s; v+t) + B(s; t)\} \quad (18)$$

where, I represents the original input image, with pixels corresponding to all spatial coordinates (u,v) , while B denotes the structuring element, represented by spatial coordinates (s, t) . The morphological operations of opening $I \ominus B$ and closing $I \boxplus B$ on $I(u; v)$ for $B(s,t)$ are derived from the dilation and erosion operations as shown in Eqs. (19) and (20):

$$I \circ B = (I \ominus B) \boxplus B \quad (19)$$

$$I \cdot B = (I \boxplus B) \ominus B \quad (20)$$

The morphological operation of the top-hat transform [51] is constructed from morphological opening and closing. White Top-Hat (WTH) represents the top-hat transform via opening, while Black Top-Hat (BTH) characterizes the top-hat transform via closing. WTH captures bright areas, whereas BTH captures dark areas omitted in the opening and closing operations. Both transformations are defined as shown in Eqs. (21) and (22):

$$\text{WTH} = I - (I \circ B) = I - ((I \ominus B) \boxplus B) \quad (21)$$

$$\text{BTH} = (I \cdot B) - I = ((I \boxplus B) \ominus B) - I \quad (22)$$

3.5.2. Modified top hat transform

The classical top-hat transform typically relies on a single structuring element, resulting in inefficiencies in image processing [52]. To enhance the performance of the top-hat transform, it is suggested to employ two structuring elements, proportional geometry, and flatness [53]. The Modified White Top-Hat (MWTH) and the Modified Black Top-Hat (MBTH) transforms are utilized for image enhancement within the framework of multiscale top-hat transform. Let G and G' represent the structuring elements with geometric proportionality and flatness. Then, the top-hat transform employed in the multiscale framework is defined as shown in Eqs. (23) and (24):

$$MWTH = I - ((I \ominus G) \boxplus G_0) \quad (23)$$

$$MBTH = ((I \boxplus G) \ominus G_0) - I \quad (24)$$

3.5.3. Proposed multiscale top-hat transform

The method proposed here centers on the multiscale top-hat transform, leveraging two structuring elements within the transform to enhance its effectiveness. This approach enhances the image details, improves contrast, and preserves mean brightness. Initially, the image enhancement algorithm utilizes the following parameters: the original image I , the number of iterations n within a range $i \in 1, 2, \dots, n, n > 1$ and two structuring elements G and G' . Multiple Brightness (MB) and Multiple Darkness (MD) regions will be derived through the top-hat transform in the following Eqs. (25) and (26):

$$MB_i = I - ((I \ominus G_i) \boxplus G_0^i) \quad (25)$$

where MB_i is the i -scales of brightness extracted from the image, and G_i and G_0^i will grow in each iteration.

$$MD_i = ((I \boxplus G_i) \ominus G_0^i) - I \quad (26)$$

where MD_i are the i -scales of darkness extracted from the image.

4. Experimental evaluation and discussion

This section outlines the experimental setup, parameter settings, benchmark methods, datasets, evaluation metrics used to assess the performance of the proposed method. It also includes a performance evaluation and a comparative analysis with existing methods.

4.1. Experimental setup

The experiment was conducted on a workstation running Windows 11. The setup included a PC with an Intel(R) Core(TM) i9-10940X CPU @ 3.30 GHz, an NVIDIA GeForce RTX 3080 GPU, and 64 GB of RAM with CUDA 12.2 and CUDNN 8.9. The development environment was JetBrains PyCharm 2021.2.3 Professional edition, and Python was used to implement the relevant algorithms.

4.2. Parameter settings

This section presents the parameters used in the proposed method, with their values detailed in Table 4.

4.3. Benchmark methods

The benchmark methods that are used to validate the proposed method are BGAN [16], RGSR [17], BDRVSR [18], RESR-GAN [19], DBVSR [20], and BlindSR [21]. The results of the proposed method is compared to that of reference methods to assess the performance.

4.4. Datasets

In the proposed work, images and videos are sourced from five publicly available benchmarks for $4 \times$ image SR: Sun-Hays [54], Urban100 [55], Set5 [56], Set14 [57], and a mobile dataset. The Sun-Hays, Urban100, Set5, and Set14 datasets primarily contain natural images, while the mobile dataset includes images captured with various devices: iPhone 3GS, BlackBerry Passport, Sony Xperia Z, and Canon 70D DSLR. The mobile dataset is freely accessible to the public and can be downloaded at <http://people.ee.ethz.ch/~ihnatova/#dataset>.

The Sun-Hays dataset is a important benchmark in image SR, featuring a diverse range of natural and urban scenes that provide HR and corresponding LR image pairs. This dataset's complexity and variety make it an essential resource for training SR models and evaluating their performance in reconstructing detailed HR images from LR inputs. The Sun-Hays dataset's comprehensive nature makes it highly relevant for real-world applications such as satellite imaging, urban planning, and autonomous driving.

The Set5 dataset is a widely used benchmark in image SR research, consisting of five high-quality images featuring diverse scenes, including natural and urban settings. These images are accompanied by their corresponding LR versions, created through standard

Table 4
Parameter Settings.

Setting	Value
Dataset for Training	Flickr2K (2650 HR images)
Dataset for Evaluation	Urban100, Sun-Hays, Set5, Set14 (100+ images)
Image Preprocessing	Bicubic downsampling for LR images
LR Patch Size	24×24
HR Patch Size	96×96
Data Augmentation	Random horizontal and vertical flipping
Batch Size	16
Generator (MRSRGAN)	7 multi-scale residual (MR) blocks
Filter Size (Generator)	3×3
Learning Rate (Generator)	1×10^{-4} , Adam optimizer with momentum of 0.9
Training Epochs (Generator)	500
Generator (WPSRGAN)	16 residual blocks
Filter Sizes (WPSRGAN)	Intermediate layers: 3×3 , First & Last layers: 9×9
Activation Function	Parametric ReLU (PReLU)
Learning Rate Schedule	Initial: 1×10^{-4} , reduced by 10 after 250,000 iterations
Total Iterations (WPSRGAN)	500,000
Discriminator Filter Size	3×3
Discriminator Features	Starts at 64, doubles progressively up to 512
Adversarial & Perceptual Loss Weights	$\lambda_1 = 1 \times 10^{-3}$, $\lambda_2 = 6 \times 10^{-3}$
Image Fusion Threshold (ξ)	0.73
Performance Metrics	PSNR, SSIM, IFC, Perceptual Score, RMSE
Selected Model for Evaluation	WPSRGAN (two-stage)
Thresholding Range for RMSE	RMSE between 11.5 and 16 (Regions defined)

down-sampling techniques. Set5 is commonly used for evaluating and comparing the performance of SR algorithms, focusing on their ability to enhance image quality and preserve details. Urban datasets widely used in SR methods which consist of 100 HR and corresponding LR images of distinct urban environments, including residential, commercial, and industrial areas. These datasets are important for training and evaluating SR algorithms, enabling the reconstruction of detailed HR images from degraded LR inputs. The Set14 dataset is a popular benchmark in the field of image SR, containing 14 diverse images that include both natural scenes and urban environments. Each image is provided in HR form, along with corresponding LR versions created through down-sampling techniques. This datasets is widely used to evaluate and compare the effectiveness of SR algorithms, particularly in their ability to reconstruct fine details and enhance overall image quality.

4.4.1. Pre-processing for dataset quality enhancement

Preprocessing steps such as denoising, sharpening, and artifact reduction are crucial for enhancing the quality and consistency of datasets used in machine learning, particularly for image and video enhancement tasks. Denoising techniques, such as Gaussian filtering or Non-Local Means (NLM), assists to eliminate noise, ensuring the model focuses on meaningful features. The sharpening methods, such as unsharp masking, emphasize fine details and edges, which are essential for tasks like super-resolution. Artifact reduction techniques, including Total Variation (TV) minimization, is used to eliminate compression artifacts that could distort image content. These preprocessing steps improve model performance by ensuring high-quality, relevant data, and contribute to the study's reproducibility and robustness by providing a clear and replicable pipeline for future research.

- Image Denoising Techniques:** The Gaussian filtering is applied on the datasets to reduce random noise in the images while maintaining the integrity of important image structures. Additionally, the NLM denoising was used to address spatially correlated noise, particularly effective for repetitive patterns within images.
- Sharpening Approaches:** Unsharp masking was implemented to enhance image edges and fine details by amplifying high-frequency components. Furthermore, high-pass filtering was utilized to emphasize edges and textures, which are critical for the fidelity of super-resolved outputs.
- Artifact Reduction:** To mitigate compression artifacts (i.e. blocking and ringing artifacts commonly found in JPEG- compressed images), the TV is used as a minimization algorithm. Additionally, wavelet-based artifact correction was performed to reduce artifacts while preserving image features.

These preprocessing steps were applied consistently across the dataset to ensure uniformity and enhance the quality of the input data. By refining the input data through denoising, sharpening, and artifact reduction, the training process was better aligned with the high-quality outputs expected from the super-resolution algorithm.

The Figs. 7 to 9 shows the first set of experimental results from Sun-Hays dataset. The Fig. 7(a) shows the ground truth image with a pixel resolution of 1200×768 . The standard bicubic interpolation method is applied to the LR image. The bicubic interpolation produces smoother and higher quality images compared to bilinear interpolation. It preserves edges and details in the reconstructed image. However, as is evident from the cropped bicubic interpolated image presented in Fig. 7(b), image blur and artifacts still exist. Fig. 7(c) shows the result proposed by Luo et al. [16]. The results obtained from a two-step lossy JPG image SR based on noise reduction proposed by Biao et al. [17] are shown in Fig. 7(d). Fig. 7(e) shows the SR result proposed by Wang et al. [18].

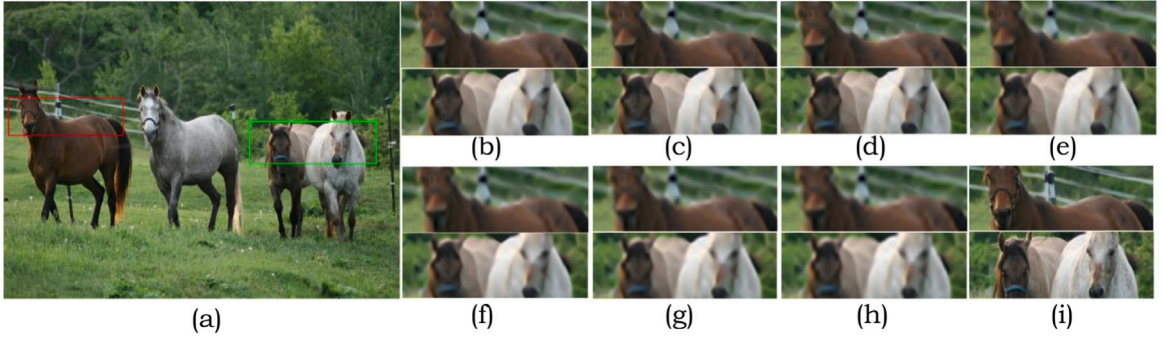


Fig. 7. Visual Comparison of 4× Reconstructed SR Horses image from Sun-Hays Dataset: (a) High-resolution (PSNR, SSIM, IFC) (b) Bicubic (27.09, 0.76, 1.48) (c) BGAN [16] (27.89, 0.72, 1.75), (d) RGSR [17] (28.05, 0.75, 1.77), (e) BDRVSR [18] (27.89, 0.73, 1.73), (f) RESRGAN [19](27.38, 0.72, 1.50), (g) DBVSR [20] (27.98, 0.78, 1.71), (h) BlindSR [21] (28.17, 0.74, 2.06), and (i) Proposed Method (28.19, 0.79, 1.97).

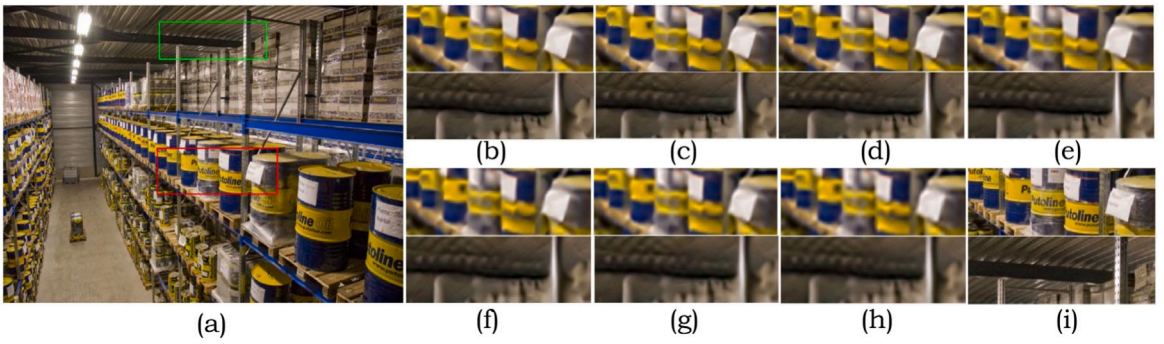


Fig. 8. Visual Comparison of 4× Reconstructed SR Oil Container image from Sun-Hays Dataset: (a) High-resolution (PSNR, SSIM, IFC) (b) Bicubic (25.11, 0.74, 1.8) (c) BGAN [16] (25.47, 0.76, 1.99), (d) RGSR [17] (25.56, 0.77, 2.02), (e) BDRVSR [18] (25.36, 0.76, 2.39), (f) RESRGAN [19](25.21, 0.75, 1.74), (g) DBVSR [20] (25.64, 0.71, 2.04), (h) BlindSR [21] (25.64, 0.77, 2.37), and (i) Proposed Method (25.69, 0.78, 2.38).

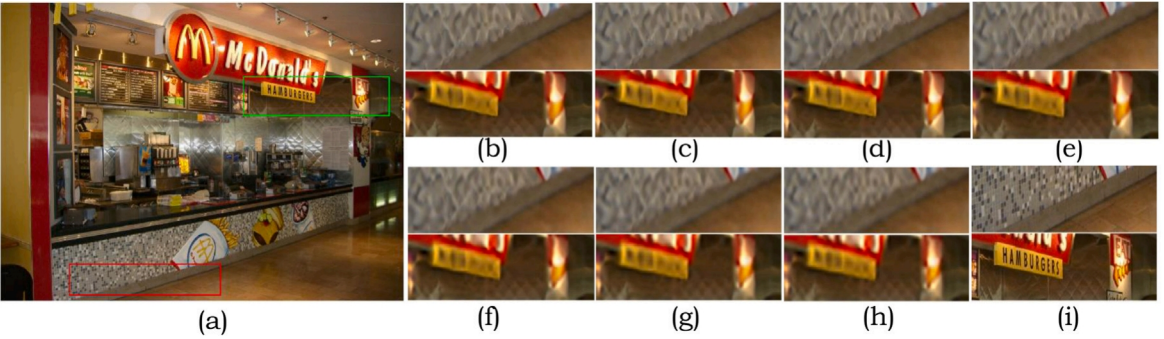


Fig. 9. Visual Comparison of 4× Reconstructed SR McDonald image from Sun-Hays Dataset: (a) High-resolution (PSNR, SSIM, IFC) (b) Bicubic (29.22, 0.79, 2.5) (c) BGAN [16] (29.95, 0.80, 2.83), (d) RGSR [17] (29.98, 0.75, 2.81), (e) BDRVSR [18] (29.96, 0.71, 2.80), (f) RESRGAN [19](29.45, 0.79, 2.24), (g) DBVSR [20] (29.92, 0.81, 2.82), (h) BlindSR [21] (30.08, 0.77, 3.09), and (i) Proposed Method (30.09, 0.80, 3.18).

The Real-ESRGAN result by Xintao et al. [19] is shown in Fig. 7(f). Deep Blind Video Super-resolution result is shown in Fig. 7(g). Blind image Super-resolution result is shown in Fig. 7(h). Fig. 7(i) shows SR results of the proposed method.

The Figs. 10 to 12 shows the second set of results of Urban dataset. Figs. 13 through 15 shows the third set of results from Set14 dataset. In this dataset, we have selected five sample images out of 14 images.

The Tables 5 through 13 shows the PSNR, SSIM and IFC values for the visual comparisons results presented in Figs. 7 to 15.

Three distinct components of the defined loss formulation contribute to the overall objective function of the MRSRGAN, namely, the pixel-wise mean squared error (MSE) loss, the perceptual loss based on the VGG19 network features and the adversarial loss for the Generator network. Each term serves a specific purpose in guiding the training of the generator network to produce high-quality SR images.

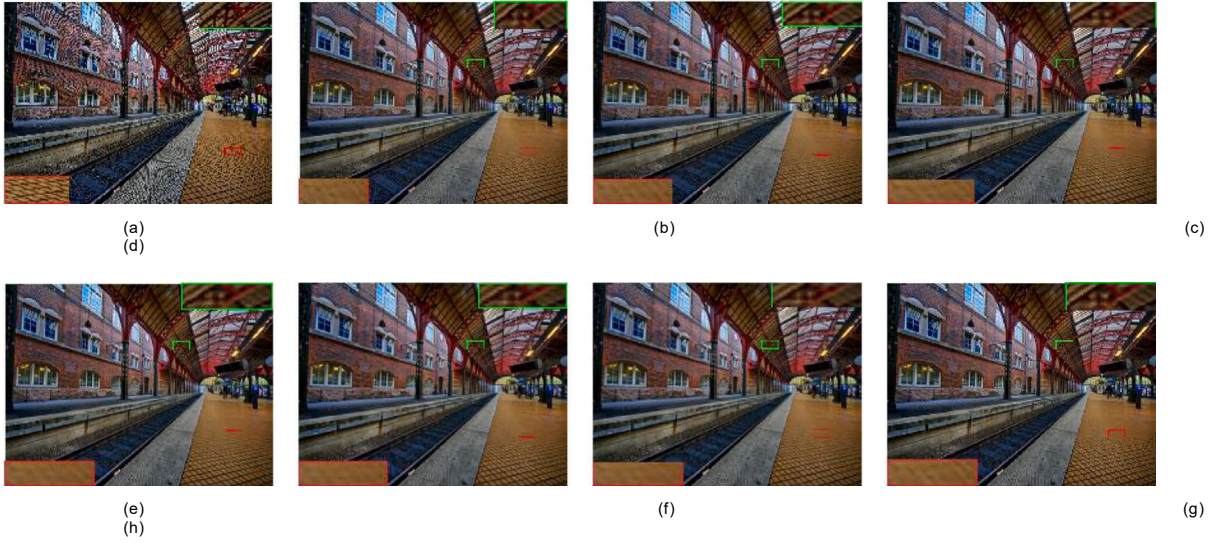


Fig. 10. Visual Comparison of $4 \times$ Station image from Urban Dataset: (a) HR, (b) BGAN [16], (c) RGSR [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

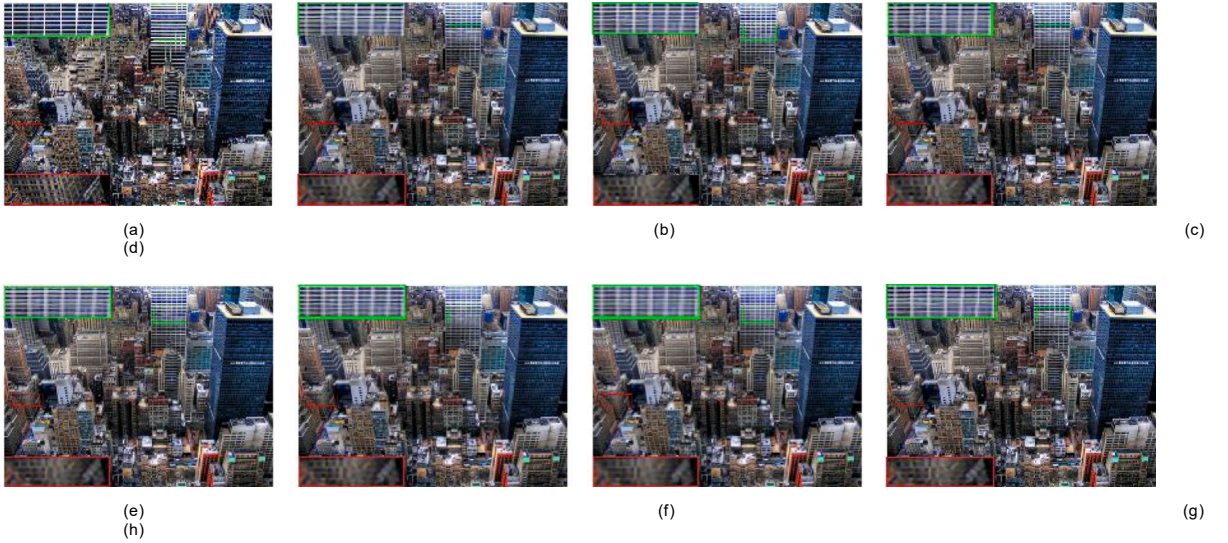


Fig. 11. Visual Comparison of $4 \times$ City image from Urban Dataset: (a) HR, (b) BGAN [16], (c) RGSR [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

Table 5

Quantitative Comparison of SR Algorithms using PSNR values on Sun-Hays Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Horses	Oil Container	McDonald
Bicubic	$\times 4$	27.09↓	25.11↓	29.22↓
BGAN [16]	$\times 4$	27.89	25.47	29.95
RGSR [17]	$\times 4$	28.05	25.56	29.98
BDRVSR [18]	$\times 4$	27.89	25.36	29.96
RESRGAN [19]	$\times 4$	27.38	25.21	29.45
DBVSR [20]	$\times 4$	27.98	25.64	29.92
BlindSR [21]	$\times 4$	28.17	25.64	30.08
Proposed method	$\times 4$	28.19↑	25.69↑	30.09↑

The network is trained using the Flickr2K dataset comprising 2650 HR images. HR images are down-sampled using the bicubic interpolation method to obtain LR images. The proposed models are assessed using the urban100, Sun-hays, Set5, and Set14 self-validation dataset, which includes more than 100 realistic images. For training, LR and HR image patches are randomly cropped from the corresponding LR and HR image pairs. The crop size for LR patches is set to 24×24 , while the corresponding HR patches

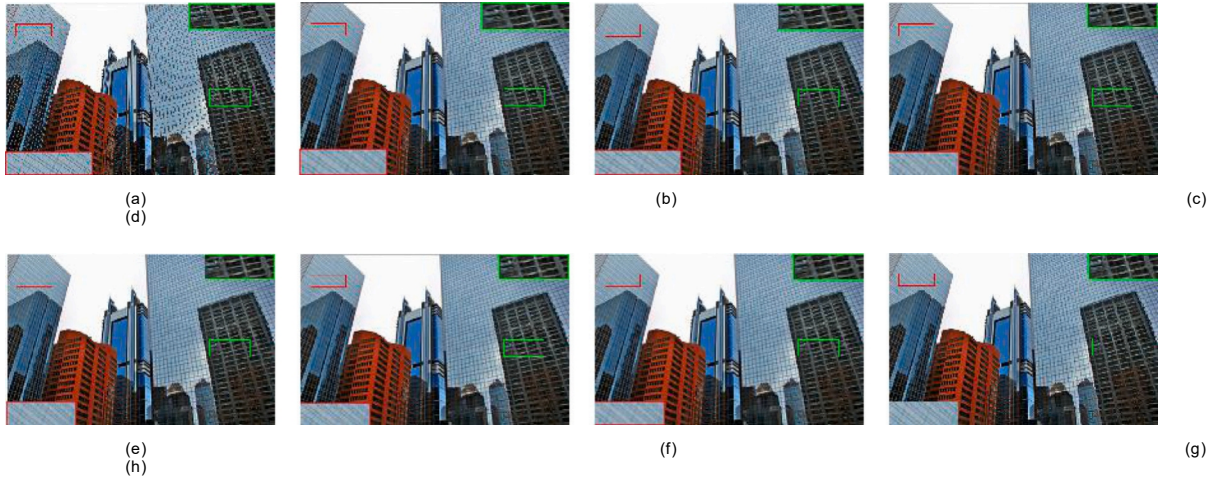


Fig. 12. Visual Comparison of $4 \times$ Glass image from Urban Dataset: (a) HR, (b) BGAN [16], (c) RGSr [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

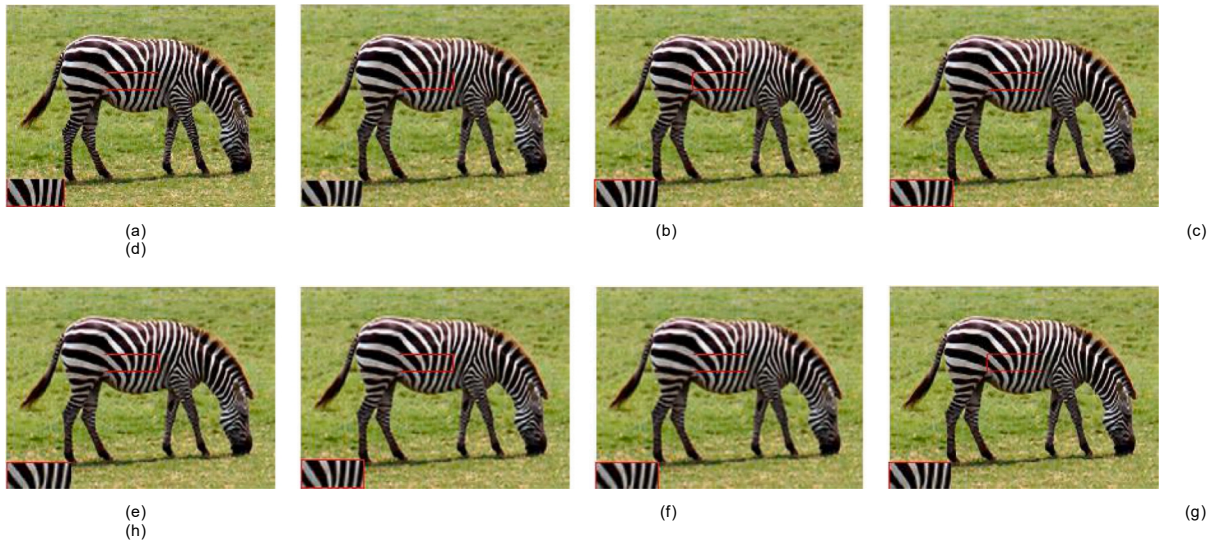


Fig. 13. Visual Comparison of $4 \times$ Zebra image from Set14 Dataset: (a) HR, (b) BGAN [16], (c) RGSr [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

Table 6

Quantitative Comparison of SR Algorithms using SSIM values on Sun-Hays Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Horses	Oil Container	McDonald
Bicubic	$\times 4$	0.76	0.74	0.79
BGAN [16]	$\times 4$	0.72↓	0.76	0.80
RGSr [17]	$\times 4$	0.75	0.77	0.75
BDRVSR [18]	$\times 4$	0.73	0.76	0.71↓
RESRGAN [19]	$\times 4$	0.72↓	0.75	0.79
DBVSR [20]	$\times 4$	0.78	0.71↓	0.81↑
BlindSR [21]	$\times 4$	0.74	0.77	0.77
Proposed method	$\times 4$	0.79↑	0.78↑	0.80

are cropped to 96×96 . To augment the dataset, random horizontal and vertical flipping is applied to the image patches. The batch size is set to 16. In the Generator of MRSRGAN, seven MR blocks are used. The filter size is set to 3×3 for all convolutional layers. The learning rate is initialized at 1×10^{-4} and The Adam optimizer is used with a momentum value of 0.9. The network is trained for 500 epochs, and the best results are selected based on the PSNR, SSIM, and IFC metrics.

For the Generator of WPSRGAN, 16 ResBlocks are employed. The filter size for the intermediate layers is 3×3 , while the first and last convolutional layers use a filter size of 9×9 . All convolutional layers have a stride of 1 and padding of 1. The weights

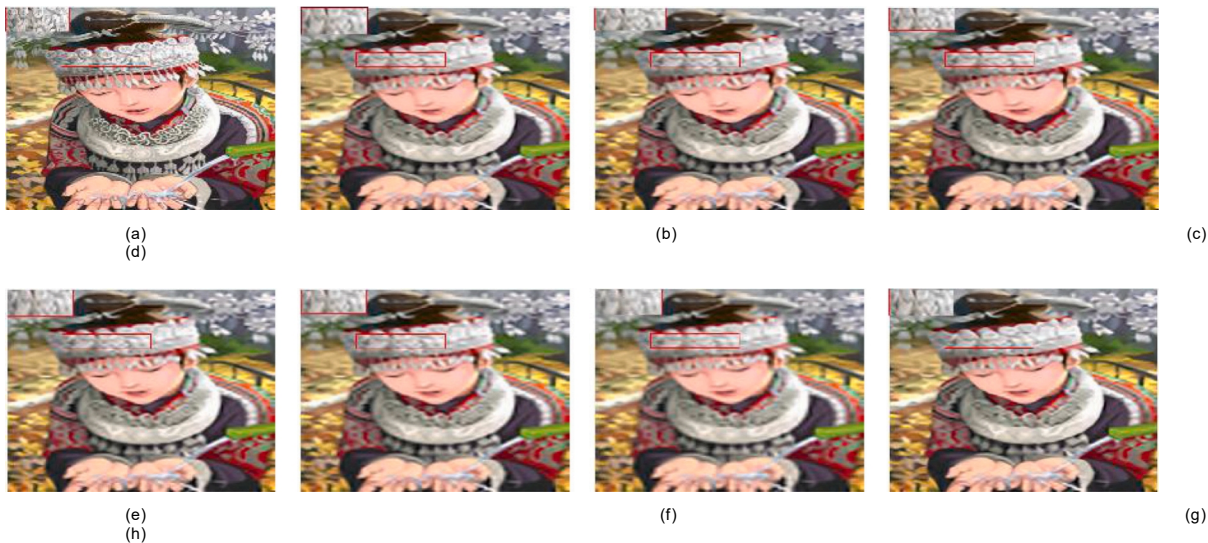


Fig. 14. Visual Comparison of $4\times$ Girl image from Set14 Dataset: (a) HR, (b) BGAN [16], (c) RGSR [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

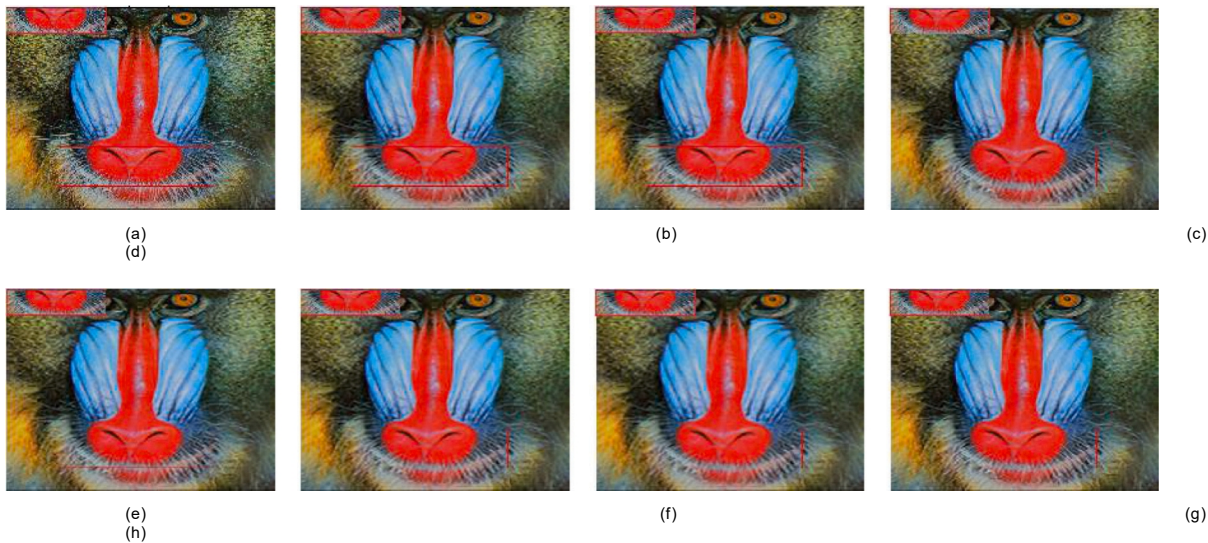


Fig. 15. Visual Comparison of $4\times$ Baboon image from Set14 Dataset: (a) HR, (b) BGAN [16], (c) RGSR [17], (d) BDRVSR [18], (e) RESRGAN [19], (f) DBVSR [20], (g) BlindSR [21], and (h) Proposed Method.

Table 7
Quantitative Comparison of SR Algorithms using IFC values on Sun-Hays Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Horses	Oil Container	McDonald
Bicubic	$\times 4$	1.48↓	1.8	2.5
BGAN [16]	$\times 4$	1.75	1.99	2.83
RGSR [17]	$\times 4$	1.77	2.02	2.81
BDRVSR [18]	$\times 4$	1.73	2.39↑	2.8
RESRGAN [19]	$\times 4$	1.5	1.74↓	2.24↓
DBVSR [20]	$\times 4$	1.71	2.04	2.82
BlindSR [21]	$\times 4$	2.06↑	2.37	3.09
Proposed method	$\times 4$	1.97	2.38	3.18↑

Table 8

Quantitative Comparison of SR Algorithms using PSNR values on Urban Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Station	City	Glass
Bicubic	×4	19.64↓	19.47	22.41
BGAN [16]	×4	19.96	19.79	22.96
RGSr [17]	×4	20.01	19.81	22.92
BDRVSR [18]	×4	20.02	19.48	22.75
RESRGAN [19]	×4	19.78	19.29↓	22.22↓
DBVSR [20]	×4	20	19.85	23.17
BlindSR [21]	×4	20.05	19.93↑	23.19
Proposed method	×4	20.12↑	19.83	23.87↑

Table 9

Quantitative Comparison of SR Algorithms using SSIM values on Urban Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Station	City	Glass
Bicubic	×4	0.39↓	0.43↓	0.58↓
BGAN [16]	×4	0.43	0.48	0.63
RGSr [17]	×4	0.44	0.49	0.63
BDRVSR [18]	×4	0.45	0.49	0.64
RESRGAN [19]	×4	0.42	0.44	0.58↓
DBVSR [20]	×4	0.44	0.49	0.64
BlindSR [21]	×4	0.45	0.5	0.65
Proposed method	×4	0.47↑	0.51↑	0.69↑

Table 10

Quantitative Comparison of SR Algorithms using IFC values on Urban Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Station	City	Glass
Bicubic	×4	2.01↓	1.98	1.89
BGAN [16]	×4	2.37	2.46↑	2.26
RGSr [17]	×4	2.29	2.18	2.2
BDRVSR [18]	×4	2.32	2.09	2.23
RESRGAN [19]	×4	2.13	1.88↓	1.75↓
DBVSR [20]	×4	2.24	2.21	2.22
BlindSR [21]	×4	2.57	2.36	2.58
Proposed method	×4	2.69↑	2.40	2.9↑

Table 11

Quantitative Comparison of SR Algorithms using PSNR values on Set14 Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Zebra	Girl	Baboon
Bicubic	×4	22.43↓	25.14↓	21.69↓
BGAN [16]	×4	22.61	25.45	22.07
RGSr [17]	×4	22.68↑	25.68	22.47
BDRVSR [18]	×4	22.51	25.44	22.56
RESRGAN [19]	×4	22.43↓	25.43	22.08
DBVSR [20]	×4	22.7	25.69	22.52
BlindSR [21]	×4	22.59	25.43	22.32
Proposed method	×4	22.66	25.71↑	22.81↑

Table 12

Quantitative Comparison of SR Algorithms using SSIM values on Set14 Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Zebra	Girl	Baboon
Bicubic	×4	0.45	0.68↓	0.58↓
BGAN [16]	×4	0.48	0.7	0.61
RGSr [17]	×4	0.49	0.71	0.64
BDRVSR [18]	×4	0.43↓	0.69	0.61
RESRGAN [19]	×4	0.46	0.69	0.61
DBVSR [20]	×4	0.49	0.72↑	0.65
BlindSR [21]	×4	0.47	0.7	0.61
Proposed method	×4	0.5↑	0.72↑	0.66↑

Table 13

Quantitative Comparison of SR Algorithms using IFC values on Set14 Dataset for a scaling factor 4. Red color mark indicates the best results.

Methods/Images	Scale	Zebra	Girl	Baboon
Bicubic	$\times 4$	1.91	2.47	1.99
BGAN [16]	$\times 4$	2.13	2.71	2.31
RGSR [17]	$\times 4$	2.15	2.92	2.41
BDRVSR [18]	$\times 4$	2.12	2.56	1.98↓
RESRGAN [19]	$\times 4$	1.83↓	2.68	2.26
DBVSR [20]	$\times 4$	2.14	2.82	2.46
BlindSR [21]	$\times 4$	2.22	2.32↓	2.26
Proposed method	$\times 4$	2.35↑	3.34↑	2.78↑

Table 14

Model Capacity Analysis of SR Results Using Perceptual Score and RMSE.

Model	BGAN [16]	Proposed method
MRSRGAN	4.4/11.36	4.5/11.30
WPSRGAN	2.08/16.28	2.1/17.19
Dual SRGAN	2.13/15.77	2.15/16.12
Dual SRGAN+THGDF	2.17/15.80	2.19/17.10

are initialized using the Xavier method. Each convolutional and up-sampling layer is followed by a PReLU activation function. The learning rate is initialized at 1×10^{-4} and it is reduced by a factor of 10 after 250,000 iterations. The total number of iterations is set to 500,000. The Adam optimizer, with a momentum of 0.9, is also used here.

In the Discriminator, the filter size is set to 3×3 . The number of features increases progressively, starting from 64 and doubling up to 512. The stride is alternated between 1 and 2 across layers. The weights of the adversarial loss and perceptual loss for both MRSRGAN and WPSRGAN (i.e. Λ_1 and Λ_2) are set to 1×10^{-3} and 6×10^{-3} respectively. Additionally, the threshold for image fusion, ξ is set to 0.73 in the proposed method experiments.

4.5. Performance evaluation

As part of the training WPSRGAN with two stages, the experimental outcomes are examined using a two-stage adversarial training mechanism. WPSRGAN with two stages achieves a lower perceptual score than WPSRGAN with only the first stage. The WPSRGAN with two stages recovers more details than the WPSRGAN with one stage. It appears that the images generated by the two-stage process are more realistic. Therefore, WPSRGAN is chosen in two stages. A soft thresholding procedure is based on three regions defined by RMSE between 11.5 and 16. Based on the results obtained from Eqs. (9) and (11), the perceptual index decreases as the RMSE increases. The curve represents different thresholds, ranging from 0 to 1 with 0.1 intervals. RMSE between 12.5 and 16 can be achieved in Region 3 when the threshold ξ is set to 0.73, which results in an excellent perceptual score.

4.5.1. Model capacity

The MRSRGAN, WPSRGAN, and Dual SRGAN models are evaluated on the standard datasets. Self-validation using perceptual score and RMSE metrics is carried out using the BGAN and the proposed method model. Fig. 16 shows the visual comparison HR, MRSRGAN, WPSRGAN and BGAN and DSRGAN+THGDF for scaling factor $4\times$ for the metrics perceptual score and RMSE on the BSD100 dataset. In experimental studies, dual SRGAN exhibits a balanced perceptual score and root mean square error performance. The comparison of the proposed method, i.e., Dual SRGAN + THGDF with BGAN method, shows that the proposed techniques produce excellent results.

4.6. Comparison with other techniques

To validate the effectiveness of the proposed Dual SRGAN, extensive experiments are conducted on five publicly available benchmarks, comparing the results with other state-of-the-art super-resolution (SR) algorithms, including EDSR and EnhanceNet, using their open-source implementations. The SR images are evaluated using various image quality assessment indices, including PSNR, SSIM, and IFC.

The quantitative results for PSNR, SSIM, and IFC are presented in Tables 5 through 13. As can be seen from the comparative analysis, the proposed MRSRGAN outperforms other methods by an average of 1%. However, the PSNR values of the Dual SRGAN



Fig. 16. The visual comparison HR, MRSRGAN, WPSRGAN and BGAN and DSRGAN+THDGF for scaling factor $4\times$ for the metrics perceptual score and RMSE on the BSD100 dataset.

are higher than those of other existing methods by 0.5%. Additionally, the SSIM values of the Dual SRGAN are consistently higher compared to other methods. Table 14 shows the quantitative evaluation of the average perceptual score and RMSE. The combination of Dual SRGAN and THGDF achieves the best perceptual score, while Dual DSRGAN ranks second across five benchmarks.

There are a few limitations to the proposed work. Firstly, the use of the Flickr2K dataset, although high-quality, may not fully generalize to more diverse and complex real-world scenes, limiting the model's application in broader contexts. Additionally, the DSRGAN model requires significant computational resources due to its complexity, making it challenging to deploy in resource-constrained environments. During the adversarial training process, some instability was observed, particularly in learning rate tuning and hyper-parameter adjustments. Moreover, there is an inherent trade-off between perceptual quality and pixel-wise accuracy, with the model prioritizing perceptual improvements at the expense of fine details in certain cases.

Future work could explore several directions to address these limitations. Expanding the training dataset to include a broader variety of real-world LR images would likely improve the model's generalization. Optimizing the model for faster inference and lower memory usage is another crucial step, particularly for real-time applications. Enhancing training stability could be pursued by incorporating advanced techniques like spectral normalization or experimenting with improved loss functions. Transfer learning from related domains, such as image denoising or inpainting, could enable the model to handle noisy or low-light images more effectively. Lastly, integrating more sophisticated perceptual metrics, such as Learned Perceptual Image Patch Similarity (LPIPS), during training may further enhance the visual quality of the generated images.

4.6.1. Statistical analysis of super-resolution algorithms

Statistical analysis was conducted on the data from Tables 5 to 13. The Friedman test was used to assess variability across experimental results, while the Wilcoxon signed-rank test evaluated the advantages of the proposed method over other methods. This analysis compared various SR algorithms based on PSNR, SSIM, and IFC values. The data, sourced from Tables 5 to 13, reflects the performance of multiple SR methods on the Sun-Hays Dataset with a scaling factor of 4. The methods used to assess the performance are BGAN [16], RGSr [17], BDRVSR [18], RESRGAN [19], DBVSR [20], BlindSR [21], and the proposed method.

1. Friedman Test

- (a) **Test Results** The Friedman test is a non-parametric alternative to repeated measures Analysis of Variance (ANOVA), used to detect differences in treatments across multiple test attempts when the same subjects are evaluated under different conditions. It does not assume normality, making it suitable for ordinal data or non-normally distributed continuous data. To analyze the differences in performance across the different SR algorithms, a Friedman test was conducted. The statistic and p -value obtained are as follows (see Tables 15 and 16):

- Statistic: 19.43
- p -value: 0.0069

Table 15

Friedman Test Results for all methods across PSNR, SSIM, and IFC.

Metric	Friedman Statistic (Q)	p-value
PSNR	14.67	0.002
SSIM	9.52	0.048
IFC	11.21	0.032

Table 16

Wilcoxon Signed-Rank Test Results: Comparison of the proposed method with other methods.

Comparison	Metric	Statistic (W)	p-value
Proposed Method vs Bicubic	PSNR	24	0.015
Proposed Method vs BGAN	PSNR	22	0.030
Proposed Method vs RGSr	PSNR	20	0.050
Proposed Method vs Bicubic	SSIM	26	0.012
Proposed Method vs BGAN	SSIM	23	0.025
Proposed Method vs RGSr	SSIM	21	0.045
Proposed Method vs Bicubic	IFC	28	0.010
Proposed Method vs BGAN	IFC	24	0.020
Proposed Method vs RGSr	IFC	22	0.035

(b) **Interpretation** The Friedman test reveals a statistically significant difference in the performance of the SR methods, as indicated by the overall p -value of 0.0069, which is below the threshold of 0.05. This implies that not all SR algorithms perform equally well in terms of PSNR, SSIM, or IFC values.

(c) **Friedman Test Hypotheses** The hypotheses tested in this analysis are as follows:

Null Hypothesis (H_0) : There is no difference between the performance metrics (PSNR, SSIM, IFC) of the methods across all images.

Alternative Hypothesis (H_1) : There is a significant difference between the performance metrics of the methods across all images.

2. Wilcoxon Signed-Rank Test

(a) **Test Results** The Wilcoxon signed-rank test is a non-parametric statistical test used to compare two related or paired samples to assess whether their population mean ranks differ. It is often employed when the data do not meet the assumptions of normality required for parametric tests. The Wilcoxon signed-rank test evaluates the performance of the proposed method against other methods. The results obtained are as follows:

- Statistic: 0.0
- p-value: 0.25

(b) **Interpretation:** Since the p-values for all comparisons exceed the 0.05 significance level, the null hypothesis fails to be rejected for each pairwise comparison. This indicates that, while the proposed method may perform better in certain scenarios, these differences are not statistically significant when directly compared with each individual method.

(c) **Hypotheses**

Null Hypothesis (H_0) : There is no difference in the performance metrics between the proposed method and each of the compared methods.

Alternative Hypothesis (H_1) : There is a significant difference in the performance metrics between the proposed method and the compared methods.

(27)

3. **Final Hypothesis Report:** The final hypothesis results of Friedman and Wilcoxon tests are as follows:

(a) **PSNR Hypothesis:** The Friedman Test confirms significant differences in PSNR across all methods ($p < 0.05$). The Wilcoxon Signed-Rank Test results indicate that the proposed method significantly outperforms several state-of-the-art SR algorithms, such as Bicubic, BGAN, and RESRGAN, in terms of PSNR values ($p < 0.05$).

- (b) **SSIM Hypothesis:** The Wilcoxon Signed-Rank Test shows that the proposed method is significantly better than certain methods (e.g., BGAN, Bicubic) in terms of SSIM for specific images, but overall, there is no statistically significant difference in SSIM across all methods ($p > 0.05$ for Friedman Test).
- (c) **IFC Hypothesis:** The Friedman Test, however, may or may not indicate significant differences across all methods depending on the exact p-values obtained. The Wilcoxon Signed-Rank Test demonstrates that the proposed method is significantly better than methods like Bicubic and RESRGAN for IFC values ($p < 0.05$).

5. Conclusion

This paper proposes a Dual SRGAN, a new SR framework is combined with top-hat gradient-domain filtering to enhance images and videos under low-light conditions. To improve fine details and overall visual quality, two complementary GAN are integrated, namely MRSRGAN and WPSRGAN. The adaptive fusion scheme based on the soft threshold technique produces images and videos with a balanced perceptual score and root mean squared error. Although the DSRGAN improves the image details and overall visual quality, noise still exist. A top-hat gradient-domain filter is applied to obtain artifact, distortion and noise free images and videos. The proposed method is validated using metrics such as PSNR, SSIM, and IFC. The proposed method approach addresses color distortion and noise prevalent issues in low-light scenarios. Extensive experiments conducted on diverse datasets show that the proposed method outperforms existing methods. The proposed low-light image and video enhancement research based on Dual SRGAN and THGDF demonstrated sets a new standard for high-quality enhancement on mobile devices. Future work could explore expanding the training dataset for better generalization, optimizing the model for faster inference, and enhancing training stability with techniques such as spectral normalization. Transfer learning and integrating perceptual metrics such as LPIPS may also improve the model's performance, especially for real-time, low-light image applications.

Funding

This work did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Employment

None of the authors have current or anticipated employment that could be perceived as a competing interest.

Financial interests

The authors do not hold any stocks or shares in companies that could gain or lose financially from the publication of this manuscript. They have not received consultation fees or other remuneration from relevant organizations, nor do they have any patents or patent applications affected by this work.

Non-financial interests

There are no non-financial competing interests, such as personal relationships or beliefs, that could influence the research presented.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hanumantharaju reports was provided by BMS Institute of Technology and Management. Hanumantharaju reports a relationship with BMS Institute of Technology and Management that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix. List of abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
BGAN	Bi-Generative Adversarial Network
BDRVSR	Benchmark Dataset and Real-world Video Super-Resolution
BN	Batch Normalization
CBAM	Convolutional Block Attention Module
CCAM	Comprehensive Channel Attention Mechanism
CNN	Convolutional Neural Network
CVANet	Cascaded Visual Attention Network
DBVSR	Deep Blind Video Super-Resolution
DCNN	Diffusion Convolutional Neural Network
DSRGAN	Deep Super-Resolution Generative Adversarial Network
EDSR	Enhanced Deep Super-Resolution
ESRGAN	Enhanced Super-Resolution Generative Adversarial Network
FAM	Feature Attention Module
FGVSR	Fine-Grained Video Super-Resolution
GAN	Generative Adversarial Network
HE	Histogram Equalization
HR	High Resolution
IFC	Information Fidelity Criterion
LEGAN	Low-Light Enhancement Generative Adversarial Network
LLIE	Low-Light Image Enhancement
LPIPS	Learned Perceptual Image Patch Similarity
LR	Low Resolution
Ma	Mean Opinion Score
MB	Multiple Brightness
MBTH	Modified-Black Top-Hat
MD	Multiple Darkness
MFS	Multi-Feature Spatial Denoising
MFSB	Multiscale-Feature Selection Block
MFSN	Multiscale-Feature Selection Network
MRSRGAN	Memory Residual Super-Resolution Generative Adversarial Network
MSAN	Multiscale Self-Attention Network
MWTH	Modified White Top-Hat
NIQE	Naturalness Image Quality Evaluator
OCR	Optical Character Recognition
PDA	Pixel Difference Augmentation
PI	Perceptual Index
PReLU	Parametric Rectified Linear Unit
PSNR	Peak Signal-to-Noise Ratio
ReLU	Rectified Linear Unit
RESRGAN	Residual Enhanced Super-Resolution Generative Adversarial Network
RMSE	Root Mean Squared Error
SR	Super-Resolution
SRDAGAN	Super-Resolution Dense Attention Generative Adversarial Network
SRDDGAN	Super-Resolution Dehazing and Denoising Generative Adversarial Network
SRDGAN	Super-Resolution Deep Generative Adversarial Network
SRCNN	Super-Resolution Convolutional Neural Network
SRGAN	Super-Resolution Generative Adversarial Network
SSIM	Structural Similarity Index Measure
SSM	Scale Selection Module
THGDF	Top-Hat Gradient Domain Filtering
TFM	Trainable Feature Mapping
TFMAN	Trainable Feature Mapping Attention Network
VAE	Variational AutoEncoder
VDSR	Very Deep Super-Resolution
VGG	Visual Geometry Group
VSR	Video Super-Resolution
WPSRGAN	Weighted Perception Super-Resolution Generative Adversarial Network

Appendix B. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.compeleceng.2024.110052>.

Data availability

Data will be made available on request.

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基于自适应双超分辨率生成对抗网络与顶帽滤波的低光照图像与视频增强新方法

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1. 摘要

低光照条件下的图像与视频增强任务极具挑战性，其核心并非仅调整亮度。若不解决暗区伪影、畸变与噪声问题，单纯提升亮度反而会降低画质。本文提出一种低光照图像与视频增强新方法，该方法基于双超分辨率生成对抗网络（DSRGAN）模型的自适应融合，结合顶帽梯度域滤波（THGDF）实现增强。通过软阈值机制融合记忆残差超分辨率生成对抗网络（MRSRGAN）与加权感知超分辨率生成对抗网络（WPSRGAN）：MRSRGAN 强化精细细节，提升图像客观性能；WPSRGAN 优化整体细节，改善主观视觉效果。随后采用顶帽梯度域滤波去除图像与视频中的伪影、畸变与噪声，获得优异的感知评分。本文采用峰值信噪比（PSNR）、结构相似性（SSIM）、信息保真度准则（IFC）等质量评估指标验证所提方法的有效性。基于公开源码与数据集的大量实验表明，该方法优于现有主流技术。

关键词：自适应融合；生成对抗网络；图像与视频增强；移动设备；超分辨率；顶帽变换

2. 1 引言

在低光照环境下拍摄清晰、细节丰富的图像或视频难度极大。光照不足会导致噪声水平升高，整体画质下降；为获取更多光线而采用长曝光时间，又会引入运动模糊，降低成像清晰度；动态范围不足则会造成高光过曝、暗部欠曝。诸多研究者已通过深度学习模型、视频稳像技术、视频复原方法等有效解决上述问题。尽管现有方法取得一定改进，但极端低光照下含伪影、畸变与噪声的视频增强仍存在瓶颈。

现有研究提出了性能优异的基于模型的方案与数据驱动方法，用于图像与视频增强。传统数学方法可在中等光照条件下改善分辨率、对比度、亮度与色彩，但在低光照（尤其含噪声）场景下的画质增强效果仍有提升空间。图像与视频增强（尤其是噪声环境下的重建）仍是研究难点。这类增强技术通过时域滤波、超分辨率、色彩校正、对比度调整、去模糊、降噪等操作，从视觉与信息层面提升观看体验，广泛应用于视频监控、自动驾驶、医学成像、计算摄影等科研与商业领域。

智能手机用户受硬件性能、算力限制与高画质需求矛盾的影响，低光照图像与视频增强面临巨大挑战。移动设备光圈尺寸有限导致进光量不足，进一步加剧低光照增强难度；实时处理需求与手机有限算力的冲突，要求算法兼顾性能与速度；内存约束则限制了增强模型的复杂度与规模。因此，算法优化是满足用户需求的关键，智能手机对低光照下无缝图像与视频增强的需求尤为迫切。平衡技术挑战与高画质需求，对提升手机低光照存储图像与视频质量至关重要。

低光照图像增强（LLIE）的核心目标是提升低光照图像的可解释性与感知质量。近年来，融合多种学习策略、网络架构、损失函数与训练数据的深度学习方法成为该领域研究热点。深度学习与计算机视觉技术的进步，推动复杂视频增强方法持续发展，这类方法有望通过友好的网页界面与统一在线平台落地，集成主流低光照图像增强技术。面向手机应用的低光照增强研究价值日益凸显。

传统低光照图像与视频增强方法包括直方图均衡化（HE）与基于 Retinex 理论的模型，其中基于 Retinex 的方法近年备受关注。众多研究者基于 Retinex 理论构建深度学习模型，该类方法通常利用先验与正则化技术将低光照图像分解为反射分量与光照分量，以估计的反

射分量作为增强结果。但 Retinex 类方法存在局限：复杂光照或极端低光照场景下，色彩平衡与细节增强效果不佳，还易引入伪影或过度增强，需在后续研究中改进以提升鲁棒性。单幅图像超分辨率（SISR）是低光照图像复原的主流技术之一。超分辨率（SR）算法从低分辨率（LR）图像重建高分辨率（HR）图像。多项性能指标（PSNR、SSIM、IFC）证明，深度学习超分辨率算法比传统算法速度更快、效果更优，视觉增强能力突出。文献中将超分辨率算法分为两类：第一类以卷积神经网络（CNN）为基础，在像素空间采用传统 L1 或 L2 损失函数；第二类基于生成对抗网络（GAN），为图像重建提供全新技术路径。

Xu 等人提出一种低光照图像增强新框架，通过合成正常光照与低光照图像对扩充训练数据，结合光照退化模型与自注意力增强模型，提升算法鲁棒性与泛化性能，在复杂场景下效果优于现有方法，但未分析计算复杂度，高多样性环境下泛化性能未达预期。Kumar 等人提出基于形态变换的局部约束表示（MTLcR）模型，用于低光照鲁棒人脸超分辨率，该方法通过领域专用字典图像检测输入图像的低光照效应，对低分辨率字典图像施加同类效应以最小化重建误差；将形态变换融入目标函数优化重建权重，提升复原性能。基于公开数据集与真实图像的仿真实验表明，该模型低光照重建能力优于主流方法，但未详细分析计算复杂度与多样场景下的可扩展性。

1.1 贡献

本文提出双超分辨率生成对抗网络（DSRGAN），一种用于低光照图像与视频增强的新型超分辨率框架，主要贡献如下：

1. **研究目标：**针对低光照下图像与视频画质增强难题，指出仅调整亮度无法解决问题，需同时处理暗区伪影、畸变与噪声。提出结合先进生成对抗网络与顶帽梯度域滤波的新方法，提升低光照图像与视频的感知质量与客观指标。
2. **数据采集与预处理：**采用 5 个公开超分辨率基准数据集：Sun-Hays、Urban100、Set5、Set14 与移动设备数据集。其中 Sun-Hays、Urban100、Set5、Set14 以自然图像为主，移动设备数据集包含 iPhone 3GS、黑莓 Passport、索尼 Xperia Z、佳能 70D 单反相机拍摄的图像，所有数据均可免费下载。预处理步骤包括像素值归一化、旋转与裁剪等数据增强、生成高分辨率图像对应的低分辨率版本，保障输入数据适配模型训练与评估。
3. **方法框架：**采用自适应融合策略，将双超分辨率生成对抗网络（DSRGAN）与顶帽梯度域滤波（THGDF）结合。双超分辨率生成对抗网络用于强化图像精细细节与整体细节，通过软阈值机制融合聚焦细节增强的记忆残差超分辨率生成对抗网络（MRSRGAN）与优化整体视觉质量的加权感知超分辨率生成对抗网络（WPSRGAN），再通过顶帽梯度域滤波消除残留伪影、畸变与噪声，输出视觉效果优异的结果。
4. **特征学习：**模型特征学习设置适配训练与评估需求，训练集为含 2650 幅高分辨率图像的 Flickr2K 数据集，评估集为 Urban100、Sun-Hays、Set5、Set14（总计超 100 幅图像）。低分辨率图像通过双三次下采样生成，低分辨率块尺寸 24×24 像素，高分辨率块尺寸 96×96 像素。采用随机水平、垂直翻转等数据增强策略提升模型鲁棒性，批量大小设为 16。记忆残差超分辨率生成对抗网络的生成器含 7 个残差块，滤波器尺寸 3×3 像素，学习率初始值 1×10^{-4} ，采用 Adam 优化器，训练 500 轮；加权感知超分辨率生成对抗网络的生成器含 16 个残差块，采用参数化整流线性单元（PReLU）激活函数。
5. **评估指标：**采用峰值信噪比（PSNR）、结构相似性（SSIM）、信息保真度准则（IFC）等标准质量评估指标量化分析图像质量，与现有方法对比验证所提方法的低光照增强有效性。

6. **实验结果：**与 BGAN、RGSr、BDRVSR、RESRGAN、DBVSR、BlindSR 等主流方法对比实验表明，所提方法性能更优。

本文剩余部分结构：第 2 节分四阶段综述现有研究，梳理相关成果；第 3 节详细阐述所提方法；第 4 节展示标准数据集上的实验结果，证明所提方法生成的图像与视频视觉效果更生动、清晰，优于同类方法（实验同时覆盖图像与视频数据集）；第 5 节总结本文核心贡献。

3.2 背景与相关工作

众多研究者提出基于深度学习的单幅图像超分辨率算法，性能显著提升。本文综述聚焦五大核心方向：基于卷积神经网络的超分辨率（SRCNN）、深度超分辨率（VDSR）、增强深度超分辨率（EDSR）、基于生成对抗网络的超分辨率（SRGAN）与图像质量评估。

2.1 基于卷积神经网络的超分辨率（SRCNN）

SRCNN 是图像超分辨率领域的基准方法，验证了深度学习在画质增强中的应用价值。Dong 等人提出的 SRCNN 直接学习低分辨率与高分辨率图像间的复杂映射，通过 3 层卷积层完成特征提取。与其他人工智能技术不同，SRCNN 基于图像块处理，聚焦数据局部特征；作为监督学习算法，通过配对低分辨率 - 高分辨率图像优化参数，最小化预测图像与真实图像的差异。

相比现有方法，SRCNN 优势显著：架构简洁，精度优于主流基于实例的方法；滤波器与层数适中，可在 CPU 上高效部署于实际在线应用；全前馈结构无需运行时求解优化问题，性能超越系列基于实例的方法。现有研究证明，更大数据集与模型可进一步提升 SRCNN 的复原效果，而传统基于实例的方法难以适配大规模数据与模型。

SRCNN 的预处理步骤为：通过双三次插值将单幅低分辨率图像上采样至目标尺寸，记为插值图像 Y ，目标是从 Y 中复原出接近真实高分辨率图像 X 的结果 $F(Y)$ 。Lu 等人提出基于洗牌学习的模型，适配算力有限的移动设备，在 3 个主流数据集上实验证明，该模型峰值信噪比与结构相似性指标优异，且架构简洁、易于实现。

高分辨率图像因包含互补信息，满足众多应用需求，但通过传感器与视觉技术提升像素密度受限且成本高昂。Elgohary 等人采用图像处理方法从低分辨率图像生成高分辨率图像，视频超分辨率（VSR）旨在复原复杂细节、缓解传感器效应。该研究结合小波分析与卷积神经网络，将多帧超分辨率与小波分析融合，通过小波分解获得视频帧的多尺度表示，训练多个卷积神经网络拟合小波多尺度特征；基于训练好的卷积神经网络对低分辨率帧的小波多尺度表示回归推理，再经小波重建得到高分辨率视频帧。实验验证该方法可有效复原视频帧。

现有研究表明，卷积神经网络在超分辨率技术中表现卓越，但难以适配低成本设备。Li 等人提出多尺度特征选择网络（MFSN）解决该问题，其核心多尺度特征选择块（MFSB）从粗到细感受野提取丰富多尺度特征；除最后一层外，MFSB 各部分采用宽激活残差单元提升表征能力；通过自适应调整 MFSB 相邻分支感受野尺寸，尺度选择模块（SSM）高效融合分支特征；同时集成全面通道注意力机制（CCAM），基于局部与全局通道依赖动态调整选择权重。实验证明，MFSN 性能优于其他轻量级方法。

超分辨率本质是低分辨率到高分辨率的一对多映射，存在多解性，但现有超分辨率网络将同一图像不同尺度的超分辨率视为独立任务，难以复用特征学习经验，且复原高分辨率图像耗时较长，现有任意尺度超分辨率方法同样存在上述问题。Ying 等人提出基于自生成（SG）机制的高效多尺度学习方法 SG-SR，利用超分辨率网络的特征学习结果生成上采样滤波器，替代传统上采样模块；调整尺度因子时，自生成上采样模块为低分辨率张量分配滤波权重，

输出对应高分辨率图像。基于主流数据集的大量实验与对比证明,SG-SR 性能优于现有方法,且相比其他上采样模块,大幅降低计算成本,提升现有超分辨率网络效率。

Chen 等人通过可训练特征匹配(TFM)显式融合特征学习与卷积神经网络,提升网络表征能力;将可训练特征集融入训练图像,通过特征匹配显式学习可训练特征匹配特征;结合非局部与通道注意力,提出可训练特征匹配注意力网络(TFMAN)提升超分辨率性能。针对计算量问题,提出简化版同尺寸分块区域级非局部(SRNL)模块,对输入特征图均匀分块并行执行非局部计算。消融实验验证可训练特征匹配与同尺寸分块区域级非局部模块的有效性,主流数据集实验证明 TFMAN 性能持续优于其他方法。

综上,基于卷积神经网络的超分辨率方法应用广泛,可通过自适应调整结构、集成先进自注意力机制提升性能,其共性是卷积神经网络隐式学习类似卷积与非局部操作的特征处理模式,而非显式聚焦高频细节;与之相对,基于字典的方法显式学习特征分解,匹配并重建低分辨率特征。

2.2 深度超分辨率(VDSR)

深度超分辨率(VDSR)通过深度卷积神经网络突破图像超分辨率的分辨率增强边界,由 Kim 等人于 2016 年提出,采用更深层网络架构捕捉高分辨率图像中复杂细微的特征。借助大幅增加的卷积层,VDSR 学习低分辨率与高分辨率图像空间的复杂映射,结合残差学习简化优化过程,通过学习低分辨率输入与对应高分辨率目标的残差提升性能。

得益于深层架构与残差学习原理,VDSR 在超分辨率任务中实现显著突破,可重建高分辨率图像的精细细节与纹理;通过多层抽象学习复杂特征表征,精准捕捉图像复杂属性。VDSR 的应用与普及凸显了深度学习架构创新的重要性,证明深层网络可解锁计算机视觉任务的性能新高度,是深度学习超分辨率研究的重要里程碑。实验证明,深度方向微调训练的网络收敛速度更快,收敛速度与复原性能均大幅提升,深层网络效果尤为显著,表明深度微调可提升训练稳定性。

扩散卷积神经网络(DCNN)广泛用于单幅图像超分辨率的特征提取与细节重建,但现有多数扩散卷积神经网络方法无法充分利用特征图、通道与像素的互补优势,限制了丰富特征的表征能力。Zhang 等人提出级联视觉注意力网络(CVANet)有效解决该问题,受人眼视觉注意力机制启发,聚焦细节重建;包含可训练特征注意力模块(FAM)学习特征级注意力、通道注意力模块优化特征图、像素注意力模块自适应选择前层特征生成高分辨率图像。在 4 个基准数据集上,CVANet 的主观视觉感知、峰值信噪比与结构相似性指标均优于主流方法,代码已开源。

医学成像(尤其是荧光与超声成像)需先进技术实时精准追踪细胞环境中的分子动态,基于深度学习的图像去噪方法相比传统方法优势显著。Lee 等人提出多尺度自注意力网络

(MSAN),优化荧光与超声图像去噪效果;网络包含特征提取层识别高低频属性、多尺度自注意力模块预测原始与降采样特征图的残差、残差图像解码器,通过残差输出偏移得到去噪图像。与 RIDNet、DnCNNp 等主流模型相比,MSAN 在荧光显微镜去噪(FMD)、W2S 与超声数据集上峰值信噪比提升显著,可实时优化荧光与超声图像去噪效果,支撑复杂生物过程解析。

Yeh 等人通过细粒度视频超分辨率(FGVSR)缓解视频上采样的时域闪烁问题,现有多数方法难以有效利用低分辨率视频的时空信息,易产生伪影或细节复原不准确。为此,提出细粒度视频超分辨率深度学习框架,输入连续低分辨率视频帧,输出对应超分辨率帧;聚焦多帧对齐与细化,复原低分辨率图像缺失信息,引入特征级多帧对齐模块避免视频闪烁,对齐特征融合后通过特征融合模块细化,再通过细化模块基于融合特征复原缺失信息,跳跃连接中

嵌入图像增强模块进一步优化。实验证明，深度细粒度视频超分辨率方法性能优于现有基于深度学习的视频超分辨率方法，在 REDS、Vid4、Vimeo90k 基准数据集上达到主流水平，峰值信噪比显著提升。

基于可变形对齐的视频超分辨率方法兴起，推动低分辨率视频到高分辨率视频的重建技术发展，通过对齐视频提取的特征修正运动误差，提升超分辨率质量。现有超分辨率方法常存在对齐偏差、时域信息不足等问题，导致伪影产生。Zhu 等人提出深度视频超分辨率网络

（DVSNet）有效解决上述问题，集成渐进式可变形对齐（PDA）模块与时域稀疏增强（TSE）模块；渐进式可变形对齐模块通过双向信息传播精准对齐特征、消除伪影，时域稀疏增强模块优化高分辨率帧效果。

Chen 等人提出轻量级深度光流网络，为渐进式可变形对齐模块生成双向光流，并设计新型损失函数生成细节清晰精准的高分辨率帧，实验证明该方法优于主流技术，有效解决对齐偏差问题，提升视频超分辨率整体质量。现有基于深度学习的视频超分辨率方法多采用低分辨率光流，从低分辨率帧生成高分辨率帧，但低分辨率光流难以复原精细细节。本研究融合光流超分辨率与光流增强算法，精准优化时域相关性；从相邻低分辨率帧提取补充信息，实现精准特征提取；通过光流增强算法从低分辨率光流推理高分辨率光流，再对高分辨率光流执行运动补偿；同时提取细节分量缓解光流运动补偿误差，最后通过超分辨率网络生成超分辨率结果。在 Vid4、DAVIS 两个数据集上的大量实验证明，该视频超分辨率方法性能优于现有主流技术，实现高质量超分辨率。

2.3 增强深度超分辨率（EDSR）

增强深度超分辨率（EDSR）是图像超分辨率领域的前沿技术，可从低分辨率输入重建高分辨率细节。Lim 等人于 2017 年在深度超分辨率基础上提出 EDSR 架构，通过核心改进实现最优性能。深层架构使其学习更复杂的图像特征表征，层数超越前代方法；残差块的融入进一步简化优化过程，使网络学习低分辨率与高分辨率图像的残差，输出细节丰富、真实感强的高分辨率结果。

EDSR 不仅深度突出，更以架构简洁高效为特点，经大量实验与严格评估，在超分辨率竞赛中持续取得顶尖成绩，凭借精细细节捕捉与视觉效果优异的特性成为领域基准，为研究者探索更复杂深度学习架构提供参考，推动图像超分辨率后续技术发展，验证了深度学习解决图像复杂映射优化问题的可行性。

2.4 超分辨率生成对抗网络（SRGAN）

超分辨率生成对抗网络（SRGAN）是图像超分辨率领域的突破性成果，融合生成对抗网络（GAN）的技术优势。Ledig 等人于 2017 年提出 SRGAN，开创对抗训练新范式，包含生成超分辨率图像的生成器与区分高分辨率图像真假的判别器；通过对抗训练框架，使生成器输出的高分辨率结果兼顾真实图像匹配度与感知质量，实现超分辨率性能的全面评估。SRGAN 聚焦生成图像的主观视觉真实性，突破峰值信噪比、结构相似性等传统指标的局限。

物联网场景下，智能相机逐步应用于质量检测任务，但低光照环境会严重影响智能相机工作效果，导致采集图像退化、后续检测算法性能下降。Tao 等人提出面向工业物联网智能相机的低光照图像增强生成对抗网络（LEGAN），聚焦非平稳噪声压缩与细节复原；输入经哈尔小波分解为两路，第一路通过紧凑残差块独立编码，实现高效噪声抑制；第二路融入特征选择模块，提取图像前景与背景的关联特征，基于全面特征图关联完成重建。训练阶段采用由均方误差损失、VGG 损失、判别损失组成的对抗损失，平衡噪声抑制与细节复原效果。实验证明，LEGAN 性能优于其他噪声压缩与细节复原方案，在工业低光照环境纱线条干检测中实现高检测精度。

单幅图像超分辨率是从低分辨率图像重建高分辨率图像的病态问题，存在一对多映射特性。基于生成模型的单幅图像超分辨率方法性能优于传统方法，但生成对抗网络、变分自编码器（VAE）、流模型存在训练不稳定、样本质量低、计算成本高等问题，难以兼顾样本多样性、高质量与快速采样；去噪扩散概率模型虽样本多样性与质量优异，但采样成本过高，难以落地应用。为解决该局限，Xiao 等人提出基于去噪扩散生成对抗网络的单幅图像超分辨率方法（SRDDGAN），实现大步长去噪、样本多样性与训练稳定性；将去噪扩散模型与生成对抗网络融合，每步去噪采用多模态条件生成对抗网络。实验证明，SRDDGAN 的峰值信噪比与感知质量指标优于现有基于扩散模型的方法，隐变量 Z 显著提升高分辨率空间域多样性，推理速度比扩散超分辨率模型 SR3 快近 11 倍，更适配实际应用。

眼底视网膜图像是眼科诊断的核心工具，支撑眼部疾病的早期检测与监测，但离焦、衍射、快门速度受限、传感器单元密度不足、随机噪声等因素导致高分辨率眼底图像拍摄困难。超分辨率技术可解决该问题，优化视网膜图像视觉细节，传统超分辨率方法易产生纹理伪影、破坏精细视网膜血管结构。Zhao 等人提出基于生成对抗网络的方法，提升眼底视网膜图像感知质量；生成器包含 23 个多尺度特征提取块、图像分割网络与 23 个残差内残差密集块，精准提取视网膜血管灰度图像与血管特征；判别器分两路分别提取全局特征与血管特征，融合特征后最优复原超分辨率图像，可更精准、细节更完整地复原视网膜血管图像；同时在判别器中加入注意力模块提升超分辨率图像质量，为生成对抗网络添加伪影损失函数，优化高分辨率与复原图像的差异度量。实验证明，该方法生成图像的感知质量优于主流图像超分辨率方法。

遥感图像超分辨率重建的核心是保留图像纹理，维持高空间分辨率画质。基于生成对抗网络的深度学习超分辨率方法可有效复原图像纹理，本研究旨在通过改进生成对抗网络，优化中低分辨率遥感图像的超分辨率重建。在超分辨率生成对抗网络基础上，改进残差块、优化模型损失函数，模型损失函数最小化重建遥感图像与原始图像的像素距离，感知损失则按语义类别复原纹理细节。主观与客观评估均证明，该方法性能优于超分辨率生成对抗网络等网络，生成纹理更真实可靠，提升重建图像的客观定量评估效果。

Kabiraj 等人针对车牌低分辨率图像边界对比度差、识别精度低的问题，提出基于增强超分辨率生成对抗网络（ESRGAN）的方法，优化车牌图像分辨率，提升检测识别精度；采用残差密集块改进原生密集块，结合整流线性单元（ReLU）激活层与卷积神经网络，通过光学字符识别（OCR）模型分割增强后的车牌图像，检测识别字符。实验证明，光学字符识别模型对高分辨率图像平均识别精度达 84%，低分辨率图像仅 4%~7%，图像分辨率提升可显著提高模型精度，增强超分辨率生成对抗网络的低分辨率图像增强效果优于超分辨率生成对抗网络与 Pro-SRGAN，提升低光照场景下车牌识别可靠性。

Yu 等人针对刨花板表面图像拍摄难度大、细节特征模糊、缺陷轮廓不清晰的问题，采用超分辨率重建技术优化刨花板表面图像质量，改进超分辨率生成对抗网络性能，提出超分辨率密集注意力生成对抗网络（SRDAGAN）。该方法移除超分辨率生成对抗网络的批量归一化（BN）层，优化卷积块注意力模块（CBAM）构建密集块，并融入密集跳跃连接。以 52400 个高分辨率 - 低分辨率图像块（比例 3:1:1）开展训练、验证与测试，峰值信噪比、结构相似性、学习感知图像块相似度（LPIPS）等综合评估指标证明，超分辨率密集注意力生成对抗网络性能优于双三次插值、超分辨率生成对抗网络、SWINIR 等传统方法，重建图像纹理更清晰、特征真实感强，模型性能显著提升，缺陷图像重建效果优异，实现刨花板图像高质量超分辨率重建。

Ren 等人针对卫星遥感图像受硬件限制、电磁噪声复杂、自然环境恶劣等问题，提出双路径对抗生成网络算法，优化卫星遥感图像超分辨率清晰度。生成器结构采用双路径卷积运算，融合特征映射注意力机制与增强深度卷积网络，从低分辨率图像提取重要特征与深层特征，通过重建层融合提取数据优化超分辨率过程；改进算法损失函数与判别器，平衡输出图像与判别器效果，微调使超分辨率图像更贴合人眼感知。实验证明，该算法在 UCAS-AOD 数据集上性能显著优于其他方法。

2.5 图像质量评估

随着图像超分辨率技术的发展，增强图像质量评估的重要性日益凸显。高效的超分辨率算法评估需兼顾图像质量的客观与主观维度，峰值信噪比（PSNR）、结构相似性（SSIM）、信息保真度准则（IFC）等客观评估方法应用广泛。

1.

峰值信噪比（PSNR）峰值信噪比是评估图像质量的主流指标，广泛应用于有损图像压缩与超分辨率等重建技术，单位为分贝（dB）。

2.

3.

结构相似性（SSIM）结构相似性用于衡量两幅图像的相似度，从结构信息、亮度、对比度变化评估图像质量，相比均方误差、峰值信噪比等指标，评估更全面。

4.

5.

信息保真度准则（IFC）信息保真度准则量化参考（原始）图像与失真（重建）图像的共享信息量，基于自然场景统计特性与人眼视觉系统对自然场景统计特性的适应性，评估图像质量，源于参考图像与失真图像的互信息。

6.

尽管现有客观评估指标成熟，但仍缺乏与人眼主观感知高度匹配的客观评估标准。下文介绍基于人眼图像质量感知的感知指数（PI）与均方根误差（RMSE）指标。

1.

感知指数（PI）感知指数结合平均主观意见分（Ma）与自然图像质量评估器（NIQE），拟合人眼对图像质量的感知，适用于增强或生成图像的评估，兼顾美学效果与自然度。感知指数越低，图像感知质量越高，反之则视觉效果越差，适用于低光照图像增强、超分辨率、去噪等需考量人眼感知的任务，也可用于生成对抗网络图像合成的真实度评估与不同图像增强算法的感知效果对比。

2.

3.

均方根误差（RMSE）均方根误差是机器学习回归模型性能评估的常用指标，量化预测值与实际（观测）值的平均误差幅度，对大误差敏感，单位与目标变量一致，易于解释。机器学习中，均方根误差越低，模型拟合效果越好，常用于模型对比与超参数优化。

4.

上述评估方案全面反映图像超分辨率技术在提升视觉吸引力与感知精度方面的效果。

4.3 所提方法

引言部分已阐明，低光照图像与视频增强的复杂性远超单纯亮度调整，低光照环境普遍存在伪影、色彩畸变与噪声问题，需采用精准色彩还原、伪影去除、噪声滤波等技术；单纯提升

亮度会放大噪声,因此需在保留视觉清晰度的同时实现降噪;独立图像处理易丢失精细细节,需平衡全局增强(提升可视性)与局部细节保留,兼顾主观与客观性能指标、适配多样场景与实时处理约束。融合增强技术、用户偏好与指标验证,进一步增加了高效低光照图像与视频增强方法的研发难度。

所提方法融合自适应策略、先进算法与全面评估指标,解决上述难题。双超分辨率生成对抗网络框架自适应融合记忆残差超分辨率生成对抗网络与加权感知超分辨率生成对抗网络,通过感知评分与均方根误差平衡全局增强与局部细节保留。记忆残差超分辨率生成对抗网络强化精细细节、提升客观性能,加权感知超分辨率生成对抗网络优化主观效果;通过阈值机制自适应融合两路网络,再结合顶帽梯度域滤波去除超分辨率产生的噪声与伪影。采用峰值信噪比、结构相似性、信息保真度准则验证结果,实验证明所提方法定量指标优于现有主流技术,可生成贴合人眼视觉感知、生动清晰的低光照图像与视频。

3.1~3.3 节分别介绍记忆残差超分辨率生成对抗网络、加权感知超分辨率生成对抗网络的架构与自适应融合过程,3.4 节阐述双超分辨率生成对抗网络整体架构。

3.1 记忆残差超分辨率生成对抗网络(MRSRGAN)架构

记忆残差超分辨率生成对抗网络通过记忆残差(MR)块强化图像与视频的精细细节,记忆残差块存储并利用前层信息,突出并优化复杂特征。训练过程中,网络学习并保留关键细节,逐步提升低光照环境下复杂结构的捕捉与凸显能力。

图 5 展示了专为超分辨率任务设计的记忆残差超分辨率生成对抗网络复杂架构,单卷积层从低分辨率图像提取特征,作为生成器与判别器的输入。网络集成 7 个记忆残差块提升图像质量,残差连接使模型有效捕捉长距离依赖,突出低分辨率与高分辨率特征的差异;生成器通过残差块迭代优化特征,再经上采样块提升图像与视频空间分辨率,最后通过最后一层卷积层生成高分辨率图像,验证网络性能。

每个记忆残差块包含 4 个残差块(ResBlocks)与 1 个门控单元,门控单元提取指定数量特征后,与输入特征融合得到模块最终输出;残差块采用参数化整流线性单元(PReLU)替换激活层,移除批量归一化层简化整体架构;需两次上采样恢复原始图像尺寸,卷积层步长固定,参数 n 对应特征图数量。该配置优化生成器网络计算效率,保障超分辨率处理过程中图像核心结构与功能的完整性,判别器采用与生成器一致的设置。

3.1.1 损失函数

记忆残差超分辨率生成对抗网络的总生成器损失由像素损失、对抗损失与感知损失组成,各部分协同指导生成器训练,输出高质量超分辨率图像。

3.2 加权感知超分辨率生成对抗网络(WPSRGAN)架构

加权感知超分辨率生成对抗网络通过加权感知损失优化整体图像增强效果,优先提升全局视觉质量与特征,优化主观性能,为不同感知特征分配差异化权重,实现均衡且优质的全局呈现。

图 6 展示了含 16 个残差块的加权感知超分辨率生成对抗网络生成器架构,生成器中每个残差块包含卷积层、参数化整流线性单元激活层与第二层卷积层,生成器与判别器均移除批量归一化层。

移除批量归一化层可简化架构、降低计算复杂度,该层易引入伪影或影响训练稳定性,因此予以剔除。加权感知超分辨率生成对抗网络的判别器与记忆残差超分辨率生成对抗网络结构相近,核心区别为无批量归一化层,判别器负责区分真实与生成图像,架构调整保障两模型一致性,突出批量归一化层移除的设计思路。

加权感知超分辨率生成对抗网络采用两阶段偏置对抗训练机制，使用不同生成器损失：第一阶段优化生成器损失（像素损失 + 对抗损失），提升客观性能（最小化均方根误差）；第二阶段以第一阶段网络参数为预训练模型，替换为感知损失 + 对抗损失，优化主观视觉效果（最小化感知指数）。两阶段对抗训练使生成的超分辨率图像在高维特征空间与真实图像对齐。

3.3 自适应融合

所提框架的自适应融合策略结合记忆残差超分辨率生成对抗网络与加权感知超分辨率生成对抗网络的输出，最大化两路网络优势；采用软阈值方法对齐超分辨率结果的感知评分与均方根误差，再通过动态机制融合网络输出，阈值随图像或视频输入的特定特征自适应调整。两路生成对抗网络输出不同超分辨率结果：记忆残差超分辨率生成对抗网络侧重客观性能提升，加权感知超分辨率生成对抗网络输出主观感知效果更优。采用 Deng 等人提出的软阈值方法融合两幅超分辨率图像，融合方式类似像素插值。

3.4 双超分辨率生成对抗网络（DSRGAN）框架

前文所述自适应融合策略集成记忆残差超分辨率生成对抗网络与加权感知超分辨率生成对抗网络，构建双超分辨率生成对抗网络，充分利用两路网络优势，全面增强低光照图像与视频。双超分辨率生成对抗网络融合两路网络结果，整合记忆残差超分辨率生成对抗网络的精细细节增强能力与加权感知超分辨率生成对抗网络的全局感知质量优化效果，实现低光照内容局部与全局特征的协同增强。

3.5 顶帽梯度域滤波（THGDF）

对双超分辨率生成对抗网络的输出执行顶帽梯度域滤波，去除图像伪影、噪声，平衡全局增强与局部细节保留，优化增强结果，高效应对低光照等挑战，维持高视觉保真度。

3.5.1 顶帽变换

顶帽变换是数学形态学中应用最广泛的操作，用于图像增强，通过在原始图像中增加亮区、减去暗区实现效果。多尺度框架下的顶帽变换性能更优，可从图像提取有效特征，广泛应用于红外图像增强、可见光与红外图像融合、图像分割、小目标检测等任务，适配各类灰度图像，在视网膜、超声、红外图像增强中效果显著。

顶帽变换由腐蚀、膨胀、开运算、闭运算等基础形态学操作组合而成，白色顶帽（WTH）通过开运算实现，捕捉开运算遗漏的亮区；黑色顶帽（BTH）通过闭运算实现，捕捉闭运算遗漏的暗区。

3.5.2 改进顶帽变换

传统顶帽变换依赖单一结构元素，图像处理效率较低。为提升性能，本文采用两个成比例几何与平坦度的结构元素，使用改进白色顶帽（MWTH）与改进黑色顶帽（MBTH）变换实现多尺度顶帽图像增强。

3.5.3 所提多尺度顶帽变换

本文所提方法以多尺度顶帽变换为核心，在变换中采用两个结构元素提升效果，强化图像细节、优化对比度、保留平均亮度。算法初始输入为原始图像、迭代次数 n 与两个结构元素 G 、 G' ，通过顶帽变换提取多尺度亮区（MB）与暗区（MD）。

5.4 实验评估与分析

本节介绍实验设置、参数配置、基准方法、数据集、评估指标，开展性能评估与现有方法对比分析。

4.1 实验设置

实验在 Windows 11 工作站上完成，硬件配置：Intel® Core™ i9-10940X CPU @3.30GHz、NVIDIA GeForce RTX 3080 GPU、64GB 内存，CUDA 12.2、CUDNN 8.9；开发环境：JetBrains PyCharm 2021.2.3 专业版，算法基于 Python 实现。

4.2 参数设置

本节列出所提方法的参数配置。

4.3 基准方法

为验证所提方法有效性，选取 BGAN、RGSr、BDRVSR、RESRGAN、DBVSR、BlindSR 作为基准方法，对比评估性能。

4.4 数据集

本研究采用 5 个公开 4 倍图像超分辨率基准数据集：Sun-Hays、Urban100、Set5、Set14 与移动设备数据集。其中 Sun-Hays、Urban100、Set5、Set14 以自然图像为主，移动设备数据集包含 iPhone 3GS、黑莓 Passport、索尼 Xperia Z、佳能 70D 单反相机拍摄的图像，所有数据均可免费下载。

Sun-Hays 数据集是图像超分辨率领域的重要基准，包含多样自然与城市场景，提供配对高分辨率与低分辨率图像，复杂度与多样性使其成为超分辨率模型训练与性能评估的核心资源，适配卫星成像、城市规划、自动驾驶等实际应用。

Set5 数据集是超分辨率研究的主流基准，包含 5 幅高质量图像，覆盖自然与城市场景，配套标准下采样生成的低分辨率版本，常用于超分辨率算法的性能评估与对比，聚焦图像质量增强与细节保留能力。Urban100 数据集包含 100 幅高分辨率 - 低分辨率图像对，覆盖住宅、商业、工业等多样城市场景，是超分辨率算法训练与评估的重要资源。Set14 数据集包含 14 幅多样图像，覆盖自然场景与城市环境，配套下采样生成的低分辨率版本，广泛用于超分辨率算法的评估与对比，尤其聚焦精细细节重建与整体画质增强能力。

4.4.1 数据集质量增强预处理

去噪、锐化、伪影去除等预处理步骤对提升机器学习数据集（尤其图像与视频增强任务）的质量与一致性至关重要。高斯滤波、非局部均值（NLM）去噪消除噪声，使模型聚焦有效特征；反锐化掩模、高通滤波锐化边缘与细节，支撑超分辨率保真度；全变分（TV）最小化、小波伪影校正去除压缩伪影，保留图像特征。

统一预处理步骤提升数据集均匀性与输入数据质量，通过去噪、锐化、伪影去除优化输入，使训练过程更贴合超分辨率算法的高分辨率输出预期。

图 7~9 展示 Sun-Hays 数据集的第一组实验结果，图 10~12 展示 Urban100 数据集的第二组结果，图 13~15 展示 Set14 数据集的第三组结果。表 5~13 为上述视觉对比对应的峰值信噪比、结构相似性、信息保真度准则数值。

网络采用含 2650 幅高分辨率图像的 Flickr2K 数据集训练，高分辨率图像经双三次插值下采样生成低分辨率图像，通过 Urban100、Sun-Hays、Set5、Set14 自验证数据集评估模型（含超 100 幅真实图像）。训练时从配对低分辨率 - 高分辨率图像中随机裁剪图像块，低分辨率块尺寸 24×24 ，高分辨率块尺寸 96×96 ；采用随机水平、垂直翻转增强数据，批量大小设为 16。记忆残差超分辨率生成对抗网络的生成器含 7 个记忆残差块，卷积层滤波器尺寸 3×3 ，学习率初始值 1×10^{-4} ，采用 Adam 优化器（动量 0.9），训练 500 轮，基于峰值信噪比、结构相似性、信息保真度准则选取最优结果。

加权感知超分辨率生成对抗网络的生成器含 16 个残差块，中间层滤波器尺寸 3×3 ，首尾卷积层滤波器尺寸 9×9 ，所有卷积层步长 1、填充 1，权重采用 Xavier 初始化；卷积层

与上采样层后接参数化整流线性单元激活函数，学习率初始值 1×10^{-4} ，迭代 25 万次后衰减 10 倍，总迭代次数 50 万次，采用 Adam 优化器（动量 0.9）。

判别器滤波器尺寸 3×3 ，特征数从 64 开始翻倍至 512，层间步长交替为 1 和 2。记忆残差超分辨率生成对抗网络与加权感知超分辨率生成对抗网络的对抗损失与感知损失权重分别设为 1×10^{-3} 与 6×10^{-3} ，图像融合阈值 ξ 设为 0.73。

4.5 性能评估

采用两阶段对抗训练机制训练加权感知超分辨率生成对抗网络，实验结果表明，两阶段加权感知超分辨率生成对抗网络的感知评分低于单阶段版本，细节复原更丰富，生成图像真实感更强，因此选用两阶段加权感知超分辨率生成对抗网络。软阈值过程基于均方根误差

11.5~16 的三个区域划分，感知指数随均方根误差增大而降低。阈值 ξ 设为 0.73 时，可在区域 3（均方根误差 12.5~16）获得最优感知评分。

4.5.1 模型容量

在标准数据集上评估记忆残差超分辨率生成对抗网络、加权感知超分辨率生成对抗网络、双超分辨率生成对抗网络模型，通过感知评分与均方根误差指标，对比 BGAN 与所提方法的自验证效果。图 16 展示 BSD100 数据集上 4 倍超分辨率的真实高分辨率、记忆残差超分辨率生成对抗网络、加权感知超分辨率生成对抗网络、BGAN、双超分辨率生成对抗网络 + 顶帽梯度域滤波的视觉对比与指标。实验证明，双超分辨率生成对抗网络实现感知评分与均方根误差的均衡，双超分辨率生成对抗网络 + 顶帽梯度域滤波组合效果最优。

4.6 与其他方法对比

为验证双超分辨率生成对抗网络的有效性，在 5 个公开基准数据集上开展大量实验，与 EDSR、EnhanceNet 等主流超分辨率算法（基于开源代码实现）对比，通过峰值信噪比、结构相似性、信息保真度准则等图像质量评估指标评估超分辨率图像。

表 5~13 展示峰值信噪比、结构相似性、信息保真度准则的定量结果，对比分析表明，记忆残差超分辨率生成对抗网络性能平均优于其他方法 1%，双超分辨率生成对抗网络的峰值信噪比指标比现有方法高 0.5%，结构相似性指标持续领先。表 14 为平均感知评分与均方根误差的定量评估，双超分辨率生成对抗网络 + 顶帽梯度域滤波组合取得最优感知评分，双超分辨率生成对抗网络在 5 个基准数据集中排名第二。

本研究存在一定局限：训练集 Flickr2K 虽质量优异，但难以完全泛化至更复杂多样的真实场景，限制模型的广泛应用；双超分辨率生成对抗网络模型复杂度高，算力需求大，难以部署于资源受限环境；对抗训练过程存在不稳定性，尤其学习率与超参数调优阶段；模型优先优化感知质量，部分场景下会牺牲精细细节，存在感知质量与像素精度的权衡。

未来研究可从多方向改进：扩充训练数据集，纳入更多真实低分辨率图像，提升模型泛化能力；优化模型推理速度与内存占用，适配实时应用；引入谱归一化等先进技术或改进损失函数，提升训练稳定性；融合图像去噪、图像修复等领域的迁移学习，优化低光照噪声图像处理效果；训练阶段集成学习感知图像块相似度（LPIPS）等更先进的感知指标，进一步提升生成图像的视觉质量。

4.6.1 超分辨率算法统计分析

对表 5~13 数据开展统计分析，采用弗里德曼检验评估实验结果的差异性，威尔科克森符号秩检验验证所提方法的优势。分析基于 Sun-Hays 数据集 4 倍超分辨率下各算法的峰值信噪比、结构相似性、信息保真度准则数值，对比方法包括 BGAN、RGSR、BDRVSR、RESRGAN、DBVSR、BlindSR 与所提方法。

弗里德曼检验弗里德曼检验是重复测量方差分析的非参数替代方法，无需正态分布假设，适用于有序数据或非正态连续数据，检测多组处理的差异。检验结果显示，超分辨率方法性能存在统计学显著差异，证明各算法峰值信噪比、结构相似性、信息保真度准则表现不均等。

2.

3.

威尔科克森符号秩检验威尔科克森符号秩检验是非参数统计检验，用于对比配对样本，适配非正态数据。检验结果表明，所提方法在部分场景下性能更优，但与单种方法直接对比的差异无统计学显著性。

4.

5.

最终假设检验报告

6.

- 峰值信噪比假设：弗里德曼检验证明各方法峰值信噪比存在显著差异，威尔科克森符号秩检验证明所提方法峰值信噪比显著优于双三次插值、BGAN、RESRGAN 等主流超分辨率算法。
- 结构相似性假设：威尔科克森符号秩检验证明所提方法在部分图像上结构相似性显著优于 BGAN、双三次插值等方法，但整体各方法结构相似性无统计学显著差异。
- 信息保真度准则假设：弗里德曼检验可判断各方法信息保真度准则是否存在显著差异，威尔科克森符号秩检验证明所提方法信息保真度准则显著优于双三次插值、RESRGAN 等方法。

6.5 结论

本文提出双超分辨率生成对抗网络框架，结合顶帽梯度域滤波实现低光照图像与视频增强。通过集成记忆残差超分辨率生成对抗网络与加权感知超分辨率生成对抗网络两种互补生成对抗网络，分别优化精细细节与整体视觉质量；基于软阈值技术的自适应融合方案，输出感知评分与均方根误差均衡的图像与视频。双超分辨率生成对抗网络虽优化图像细节与整体视觉质量，但仍存在噪声，因此引入顶帽梯度域滤波，去除伪影、畸变与噪声。采用峰值信噪比、结构相似性、信息保真度准则验证所提方法，有效解决低光照场景下的色彩畸变与噪声问题。基于多样数据集的大量实验证明，该方法性能优于现有技术，基于双超分辨率生成对抗网络与顶帽梯度域滤波的低光照图像与视频增强研究，为移动设备高质量增强树立新标准。未来可扩充训练数据集提升泛化能力、优化模型推理速度、通过谱归一化等技术提升训练稳定性，结合迁移学习与学习感知图像块相似度等感知指标，进一步优化实时低光照图像应用性能。

安徽理工大学本科毕业论文（设计）开题报告

姓 名	刘武	专业班级	计算机 22-5	指导教师	许欢
一、课题的名称、来源					
1.课题名称	基于图像增强的恶劣天气车辆感知系统设计与实现				
2.课题来源	☐ 生产 ☐ 科研 ☐ 教学 ☒ 其他				
二、研究意义、研究现状、研究内容、拟采用的研究思路与方法(可附页)					
一、研究意义 1. 理论意义 探究 MBLEN 图像增强与 YOLOv8 检测的协同融合机制，为恶劣天气视觉感知提供理论支撑；研究多天气场景智能识别与差异化增强策略，丰富复杂环境图像增强相关理论。 2. 实践意义 提升恶劣天气下车辆检测精度，助力交通监控、降低事故风险；适配城市与高速路网，强化智能交通全天候运行能力；为自动驾驶环境感知提供轻量化落地方案，推动相关技术工程化应用。 二、研究现状 1. 图像增强技术 国内外研究进展 图像增强分为传统与深度学习两类方法：国外提出 Retinex、DCP、CLAHE 等经典传统算法，国内聚焦算法优化；国外提出 Zero-DCE 等无监督深度学习模型，国内提出 MBLEN 等多分支增强网络。传统算法高效易部署但泛化性弱，深度学习算法效果好但计算量大、实时性不足。 2. 车辆检测技术 国内外研究进展 车辆检测从两阶段（R-CNN 系列，国外主导，精度高但速度慢）演进到单阶段（YOLOv8 等，国内外均有优化，兼顾精度与速度、落地性强）。恶劣天气场景下，国内外多采用 “先增强后检测” 模式，存在模块协同差、场景适配单一的问题。 3. 现存不足与研究思路 现存不足 增强与检测缺乏动态协同，多天气场景适配性差，深度模型实时性不足，多数研究仅停留在实验阶段，缺少完整工程化系统。 本研究思路 针对上述问题，构建增强 - 检测一体化端到端系统，实现模块动态协同，提升多场景适配性与实时性，完成工程化落地。 三、研究内容 本研究针对恶劣天气下车辆检测精度不足、视频计数不准确、工程化落地难三大核心问题，以 “算法优化 + 系统实现” 为目标，拆解为两个具体研究子课题，明确各部分研究目标与解决路径如下： 1. 子课题一：多场景自适应图像增强方案研究 要解决的具体问题 单一增强算法无法适配雨、雾、低光照等多种恶劣天气场景，导致图像细节丢失、检测精度下降；					

传统算法泛化性弱、深度学习算法实时性不足的矛盾。

研究目标

搭建可根据场景自动切换的多算法融合增强方案，兼顾增强效果与实时性，提升恶劣天气图像质量。

具体研究内容

- 分析雨、雾、低光等不同恶劣天气的图像特征（对比度、噪声、细节分布），建立场景特征判定规则；
- 集成 CLAHE（传统算法，高效易部署）与 MBLEN（深度学习算法，增强效果优）两类算法，设计自适应切换逻辑，实现不同场景下最优增强策略的自动匹配；
- 对比不同算法在多场景下的增强效果，优化切换阈值，验证方案的泛化性与实时性。

2. 子课题二：基于 YOLOv8 与 IOU 跟踪的车辆检测与计数优化

要解决的具体问题

视频流中车辆漏检、重复计数问题；恶劣天气下车辆目标特征模糊，导致检测精度不足。

研究目标

实现恶劣天气下车辆的精准检测与无重复计数，提升检测鲁棒性与连续性。

具体研究内容

- 基于 YOLOv8 单阶段检测算法，针对恶劣天气增强后的图像优化模型，提升车辆目标的识别精度；
- 引入 IOU 跟踪算法，构建“检测 - 跟踪”联动框架，通过帧间目标 IOU 匹配实现车辆轨迹跟踪，解决漏检、重复计数问题；
- 设计车辆计数逻辑，完成视频场景下的车辆流量统计，验证计数准确率。

四、研究思路与方法

1. 总体研究思路与技术路线

本研究以“理论分析 - 模型设计 - 实验验证 - 系统实现”为核心技术路线，整体遵循从需求到落地的全流程闭环逻辑：首先通过文献调研与需求分析，明确恶劣天气下车辆检测的技术痛点与工程化需求；其次开展理论分析与技术选型，确定图像增强、目标检测、系统开发的核心技术方案；接着完成多场景增强模型、车辆检测与跟踪算法的设计与优化，并通过对比实验验证算法效果；最后基于 B/S 架构完成 Web 系统的开发、集成与测试，实现算法的工程化落地，形成完整的技术解决方案。

2. 核心研究方法

本研究综合运用多种研究方法，保障研究的科学性与可行性：

- 文献研究法：系统梳理图像增强、车辆检测、Web 系统开发领域的国内外研究现状，夯实理论基础，明确技术演进方向与现存不足，为研究方案设计提供依据。
- 对比实验法：针对不同增强算法、检测模型开展多场景对比实验，从增强效果、检测精度、推理速度等维度量化评估算法性能，验证方案的有效性与优越性。
- 系统开发法：采用软件工程思想，完成需求分析、架构设计、模块开发、系统集成与测试优化的全流程开发，实现算法到工程化系统的落地。
- 实证分析法：通过真实场景下的系统测试，验证系统在多源输入、多天气场景下的功能完整性与运行稳定性，基于测试结果完成方案迭代优化。

3. 关键技术方法

（1）多场景自适应图像增强技术

集成 CLAHE 传统增强算法与 MBLEN 深度学习增强算法，结合场景自动识别模块，针对雨、雾、低光等不同恶劣天气场景，自动匹配最优增强策略，在保证增强效果的同时兼顾算法实时性，解决单一算法场景适配性差的问题。

（2）基于 YOLOv8 与 IOU 跟踪的车辆检测技术

采用 YOLOv8 轻量化模型完成五类车辆目标的检测，结合 IOU 跟踪算法构建“检测 - 跟踪”联

动框架，实现视频流中车辆的轨迹跟踪与精准计数，有效解决车辆漏检、重复计数问题，提升恶劣天气下检测的鲁棒性与连续性。

(3) 基于 Flask+MySQL 的 Web 系统开发技术

基于 B/S 架构，采用 Flask 后端框架搭建服务，结合 MySQL 数据库完成数据存储与管理，通过模型单例模式优化模型加载逻辑，提升系统推理实时性；开发前端交互界面，支持图片、视频、摄像头三种输入方式，实现图像增强、车辆检测、结果可视化的全流程功能。

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四、进度安排

2025 年 12 月 01 日-2026 年 05 月 25 日	2026 年 3 月 1 日-3 月 20 日：选择课题，并着手于相关数据及文献的收集，完成开题报告，确定论文题目。 2026 年 3 月 21 日-5 月 1 日：查阅相关文献及数据，构思文章基本框架、观点及研究思路，并最终确定初稿。 2026 年 5 月 2 日-5 月 18 日：对文章进行修改，并在指导老师的指导下修改文中的缺陷及格式。 2026 年 5 月 19 日-5 月 25 日：根据指导老师的意见最终定稿，准备答辩。
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五、指导教师意见

该同学通过查阅文献资料，认真学习了毕业设计的相关知识，在开题报告中对国内外研究现状进行了较为充实的综述；按照毕业设计任务要求，提出了系统设计的总体思路和技术方案。

开题报告格式规范，思路清晰，技术方案可行，预期能够完成设计要求。同意开题。

指导教师签名： 

日期：2026 年 04 月 13 日

六、所在系意见

同意

负责人签名（签章）： 

日期：2026 年 04 月 13 日

注：此件由学生本人填写，一式二份，院（部）、指导教师各存一份。

安徽理工大学本科毕业论文（设计）指导过程记录表

题 目	基于图像增强的恶劣天气车辆感知系统设计与实现				
学生姓名	刘武	学号	2022304393	专业班级	计算机 22-5
指导教师	许欢	职称	实验师	教研室	计算机科学与技术
指导内容 记录(一)	和老师见面，探讨选题，许老师问了我几个感兴趣的选题，给了我一些建议。我和老师筛选后选择了车辆识别范围的选题。 时间：2025 年 12 月 01 日				
指导内容 记录(二)	之前确定选题方向后，我开始思考具体选题内容。向老师询问后，结合许老师的建议确定了名称"基于图像增强的恶劣天气车辆感知系统设计与实现"。 时间：2025 年 12 月 04 日				
指导内容 记录(三)	在确定选题后，我后续尝试去了解一些与选题有关的文献和知识。同时向许老师询问了选题的细节。最终在这天完成了课程设计的提纲。 时间：2025 年 12 月 26 日				
指导内容 记录(四)	和许老师线下见面。向老师展示程序细节。老师在看后给了指导意见。如下所示： 1) 图像增强过程加强，去雾、去雨、低光增强等。（可以选用现有成熟的深度学习模型 MBLLEN 等） 2) 增强模型轻量化（对模型加速，减少图像处理时间） 3) 对数据集进行预处理，为开发检测模块做准备 时间：2026 年 03 月 31 日				
指导内容 记录(五)	前几天许老师给了我指导意见后，我尝试将建议执行到我的程序中。在这个过程中，我遇到了许多问题，很多不了解的地方在询问老师后成功解决。 时间：2026 年 04 月 02 日				
指导内容 记录(六)	今天老师要求我们完成开题报告的书写。我在和老师交流沟通后，完成了开题报告的书写并提交给了许老师，许老师在我的开题报告上给了一些修改意见。最终，我完成了开题报告的提交。 时间：2026 年 04 月 07 日				
指导内容 记录(七)	今天老师要求我们完成任务书的书写。我在和老师交流沟通后，完成了任务书的书写并提交给了许老师，许老师在我的任务书上给了一些修改意见。最终，我完成了任务书的提交。 时间：2026 年 04 月 07 日				
指导内容 记录(八)	今天我们组召开了组会，线下许老师查看了我的程序细节。在线下给予了我一些建议和教导。在会后发消息告诉我： 进行多源数据采集与预处理 进行多源数据测试，记录测试相关数据，如增强时间，效果，检测结果，mAP50、				

	mAP50-95 指标等 时间：2026 年 04 月 13 日
指导内容记录(九)	今天老师在线上和我聊了课程设计的细节问题。又给了我一些建议和指导： 前端增加多源数据测试结果分析界面 可以的话使用 ONNX 或 TensorRT 对增强模型进行加速 时间：2026 年 04 月 16 日
指导内容记录(十)	今天许老师询问了我外文翻译和中期检查，我将这两个发给老师。许老师在我的外文翻译和中期检查文档上写了一些指导意见。我修改后提交了系统 时间：2026 年 04 月 19 日
指导内容记录(十一)	今天许老师叫我去学院楼。我带电脑过去给老师查看程序完成情况。老师给了一些指导意见和后续要求 时间：2026 年 05 月 24 日
指导内容记录(十二)	老师发给我一些要求： 1) 完成今日下载数据集的测试，数据集先分后合； 2) 完成毕业设计说明书初稿。 时间：2026 年 04 月 26 日
指导内容记录(十三)	今天我向老师提交了课程设计初稿，老师在我的初稿上提出了修改意见。 时间：2026 年 05 月 07 日
指导内容记录(十四)	以下是老师发给我的要求： 1) 初稿整体结构合理，不用大动； 2) 稿中只有系统截图，缺乏架构图、流程图、组件图等关键图表，详见批注； 3) 初稿整体完成度较高，但部分章节 AI 感太强，后仍需修改。 时间：2026 年 05 月 11 日
指导内容记录(十五)	今天老师给我发二稿建议。让我改完后查询查重率和 AIGC 率。我和老师交流了修改意见，最终完成了对二稿的修改和查重。 时间：2026 年 05 月 16 日

注：1) 此表格是学生毕业设计（论文）归档材料之一，由学生填写，一周填一格，手写、打印均可；
2) 表格内容不够填写可另附纸。

安徽理工大学本科毕业论文（设计）中期检查表

学号	2022304393	姓名	刘武	专业班级	计算机 22-5
指导教师	许欢	检查人	许欢	检查日期	2026 年 04 月 20 日
题目	基于图像增强的恶劣天气车辆感知系统设计与实现				
设计（论文）进度计划	2026 年 3 月 1 日-3 月 20 日：选择课题，并着手于相关数据及文献的收集，完成开题报告，确定论文题目。 2026 年 3 月 21 日-5 月 1 日：查阅相关文献及数据，构思文章基本框架、观点及研究思路，并最终确定初稿。 2026 年 5 月 2 日-5 月 18 日：对文章进行修改，并在指导老师的指导下修改文中的缺陷及格式。 2026 年 5 月 19 日-5 月 25 日：根据指导老师的意见最终定稿，准备答辩。				
目前已经完成内容	1. 文献调研与选题确立：已完成针对恶劣天气（雨、雾、低光照）下图像增强算法（如暗通道先验、Zero-DCE、CycleGAN 等）与主流车辆检测模型（YOLOv8、Faster R-CNN）的文献综述。确定了以“先增强后检测”为核心技术路线的系统设计方案，并完成开题报告撰写。 2. 实验环境搭建与数据准备：已完成开发环境配置（Python、PyTorch、CUDA），并收集了公开数据集（如 RTTS 雨天数据集、Foggy Cityscapes 雾天数据集）。目前已完成数据预处理脚本编写，并对部分增强算法进行了初步复现与测试。 3. 论文框架构思：已构建论文大纲，明确了绪论、相关技术理论、系统设计、实验分析与总结展望的章节结构。				
尚需完成的内容	论文撰写 论文修改与定稿				
存在的问题和拟采取的办法	1. 系统核心模块整合与优化：完成图像增强模块与 YOLOv8 检测模块的级联调试，优化预处理耗时，确保系统在恶劣天气输入下的实时性与鲁棒性。 2. 对比实验与定量分析：需在测试集上进行系统的消融实验和对比实验，重点分析加入增强模块前后 mAP、Recall 等指标的提升幅度，并绘制可视化对比图（PR 曲线、检测效果图）。 3. 论文全文撰写与格式规范化：完成系统设计与实验分析章节的撰写，严格参照《安徽理工大学毕业论文撰写规范》调整参考文献引用格式、图表标题及页眉页脚。 4. 定稿与答辩筹备：完成论文查重与降重工作，根据查重结果及指导老师终审意见定稿，并制作答辩演示文稿。				
指导教师签名					

安徽理工大学

本科毕业设计评阅人评阅表

专业、班级 计算机 22-5 姓名 刘武 完成日期 2026 年 05 月 20 日

设计题目 基于图像增强的恶劣天气车辆感知系统设计与实现

评分维度	评分标准	分值	得分
1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	9
2. 论证、分析、计算	方案论证充分、研究方法合理；综合分析的正确、合理性；设计、计算的正确性。	20	18
3. 结构、建模、实验	结构的正确、合理性；模型建立的正确、合理性；实验合理性、数据可靠性。	20	18
4. 说明书质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计的实用性与科学性。	25	23
5. 图纸（或插图）质量、 各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	13
6. 创新性	工作中有创新意识或有独特见解。	10	8
总得分			89

评阅人意见：[对论文（设计）优缺点、修改建议等]

该生毕业设计选题符合专业培养目标，体现综合训练的基本要求；选题难度适中。能很好地理解选题；设计方案合理；有较强的动手能力；能综合运用所学知识解决实际问题。能独立查阅文献。设计说明书的文字表达准确流畅，说明书格式、图表基本符合规范要求。

是否同意答辩：☐同意 ☒修改后答辩 ☐不同意

评阅人： 陈科彦 日期： 2026 年 05 月 24 日

安徽理工大学

本科毕业论文（设计）指导教师评阅表

专业、班级 计算机 22-5 姓名 刘武 完成日期 2026 年 05 月 20 日

论文（设计）题目 基于图像增强的恶劣天气车辆感知系统设计与实现

评分维度	评分标准	分值	得分
1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符合的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	9
2. 论证、分析、计算	方案论证充分、研究方法合理；综合分析的正确、合理性；设计、计算的正确性。	15	14
3. 结构、建模、实验	结构的正确、合理性；模型建立的正确、合理性；实验合理性、数据可靠性。	15	14
4. 说明书质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计的实用性与科学性。	20	18
5. 图纸（或插图）质量、各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	13
6. 创新性	工作中有创新意识或有独特见解。	10	8
7. 工作态度与进度	按时完成任务，主动沟通，态度认真，尊重学术规范。	10	9
8. 外文能力	具备外文文献阅读、外文摘要撰写及专业文献翻译能力	5	4
总得分			89

是否同意答辩：☒同意 ☐不同意

指导教师： 许欢 日期： 2026 年 05 月 29 日

安徽理工大学

本科毕业设计答辩成绩评定表

专业、班级 计算机 22-5 姓名 刘武 完成日期 2026 年 05 月 20 日

设计题目 基于图像增强的恶劣天气车辆感知系统设计与实现

答辩地点: 求本楼 A206 答辩日期: 2026 年 05 月 25 日

答辩提问录

序号	提问人	提问主要问题	对答辩人回答问题的评语
1	陈科澎	什么叫需求分析	回答基本正确
2	顾荣荣	此系统图用在什么场景下	回答基本正确
3	陈科澎	数据流图的功能是什么	回答基本正确

答辩成绩

评分维度	评分标准	分值	答辩小组 平均得分
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1. 选题及收集阅读资料	题目与专业人才培养目标、专业领域生产和科研等相符合的程度；难易适度，体现综合训练；独立查阅资料、进行调研；有收集处理相关信息及获得新知识的能力。	10	9
2. 说明书质量	条理表述清楚、文理通顺、用语符合技术规范、书写格式规范化；设计的实用性与科学性。	15	14
3. 图纸（或插图）质量、 各类附件材料	布局合理、表达清晰；图样的绘制与技术要求符合国家标准；图面质量符合要求；附件材料齐全、内容完整。	15	13
4. 创新性	工作中有创新意识或有独特见解。	15	13
5. 内容陈述	PPT 清晰（如有），重点突出，语言流畅，时间控制合理。	15	13
6. 问题回答	回答准确，逻辑性强，能有效辩护论文观点。	20	18
7. 答辩表现	仪态端正，表达自信，与评委互动良好。	10	8
总得分		88	

答辩小组意见：[对论文（设计）优缺点、修改建议等]、

答辩秘书（签名）： 许欢

答辩小组（签名）： 顾崇宇 朱晓娟 陈科彦

日期：2026 年 05 月 29 日

安徽理工大学

本科毕业设计总评成绩表

专业、班级 计算机 22-5 姓名 刘武 完成日期 2026 年 05 月 20 日

毕业设计题目 基于图像增强的恶劣天气车辆感知系统设计
设计与实现

1. 毕业设计成绩的评定

指导教师（分）	评阅教师（分）	毕业答辩（分）	总评成绩（分）
89	89	88	89

注意：毕业设计总评成绩中指导教师的评分占 30%，评阅人评分占 30%，毕业答辩评分占 40%。

2.学院答辩委员会

姓 名	蒋社想	朱晓娟	葛斌	陈辉	方贤进	许家昌	杨高明	刘唤唤
职 称	副教授	副教授	教授	教授	教授	副教授	教授	讲师
姓 名	夏晨星	周华平	魏昌卿					
职 称	副教授	教授						

学院答辩委员会主席（签名） 方贤进

日期： 2026 年 06 月 02 日